

-
-
-
-
-

Chapter 2

Building AE Processes



Before You Start

This is a tutorial. Please read this if you want to know how to build Ae1, Ae2 and Ae3 from start to finish. These are the Active Enterprise Business Processes.

However, you must be familiar with the basic functions in Constructor. If not then you will find it helpful to read the Getting Started section in the Constructor online help text and perhaps print parts of it to keep by you for reference. To do this select the Help Menu on the Constructor Menu Bar.

Important Warnings



Warning: If you are printing sections of this Help Guide from Acrobat Viewer please check that you have the correct print resolution on your machine. Please refer to the glossary definition, Acrobat Printing Problems, for instructions.



Tip: Print this page to keep by you. Right click now for Print Topic option.



Warning 1: Do not edit Properties - you will update the Design Time database for everyone.



Warning 2: Take great care with Data Fields. JBA recommend that you do not add new fields without taking advice from your JBA consultant.

Explanation For Warning 1

During the course of this tutorial you will be placing the activity symbols onto the Workspace. These symbols represent RPG programs. The descriptions and properties for these activities have been defined in the programs.

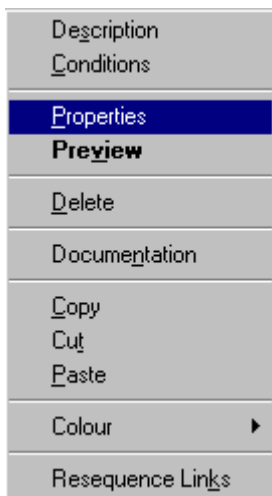
For example, a program may have been written so that the user may exit in two different ways, perhaps by OK and Cancel buttons. This would be represented by a symbol with two exit nodes which could have tooltips which say Successful and Unsuccessful.

Print
invoices



Constructor gives you the facility to go in and edit properties for activities, using a right click menu as shown below, but please don't do it because you will update that activity on the design time database and this will affect the designs for other Constructor users! This facility is here only for people who are writing their own RPG code. This tutorial will not ask you to do this.

If you drag out a symbol and you are not happy with the nodes or descriptions, as defined in its Properties, please either live with it and work round it or check that you have the correct design time database installed. If the Properties of an activity are incorrect they should be corrected on the database, not on the Workspace.



Do not edit Properties for Activities.

However, you can safely edit Manual Activities, Action Agents, Scheduling details and Control details. The tutorial will take you through step-by-step.

Explanation For Warning 2

There is a Chapter called Adding Data Fields, however, JBA recommend that you do not add new fields without taking advice from your JBA consultant.

Note that there are four types of Data Fields: Text, SQL, Programmatic and Workflow.

You can safely add Text and SQL types. However, take great with Programmatic and Workflow types as follows:

Programmatic



Warning: If you add a Programmatic type of Data Field then you need to create an RPG program with a subroutine for each Data Field. Please refer to the appendices called Variable Substitution Skeleton Program and Data Field List to help you with this.

Calls program defined in Constructor to evaluate Data Fields. All programs written to evaluate run-time Data Fields will be of the same structure as the Skeleton program QRPGRSRC/@SWMVARSUB. That is same input parameter list, output P#VARV.

Note that one program may be used to evaluate many Data Fields. Each Data Field will have its own subroutine within its program.

Workflow



Warning: Workflow Data Fields need to be supported by a Work Management program. Work Management Data Fields are directly supported by the Work Management Engine. You will only need to amend these under exceptional circumstances and with guidance from JBA.

These are the Work Management information Data Fields:

- Company
- Business Process
- Business Object
- Document
- Activity
- Current Time
- Current Date.

Set Up Users

Stage 1

For this demonstration to work you will need to setup data on the descriptions file INP15. This will be used by the Data Fields you work with in the tutorials. Please set up entries where the Primary Key is SLMN and the Secondary key is the Company Number. The data you enter is to be the User's Name.

Stage 2

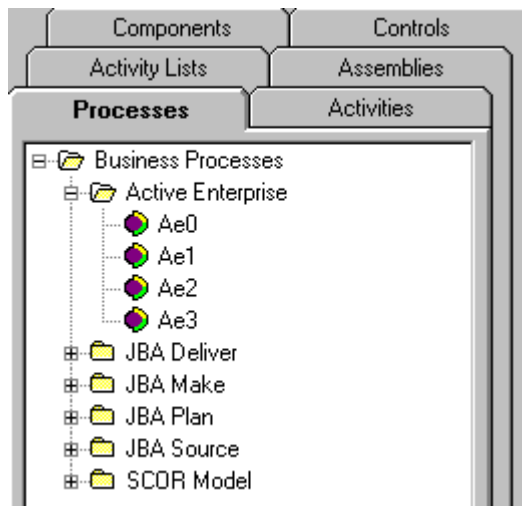
Remember to set up corresponding Users for Explorer, using the User Role Maintenance program which you select from the Program List on your Start Menu. There is a tutorial which shows you how to do this in the online Help from the User Role Maintenance main menu.

You then need to change to this user using the Change Explorer User button on the Explorer Toolbar. You can then view this user's Action List.

Active Enterprise Data

To make sure that you will be able to see the Active Enterprise data please do the following:

1. Select Preferences on the Edit Menu.
2. Click the System Configuration button.
3. Select the File Paths tab.
4. Look for the group box at the top of the window called Application Version. There is a drop down list box in here. From this list please select EPM352. If you cannot see this option then please consult the person who installed Constructor before going any further. You should have these SCOR models installed on your machine.
5. Select File and Save Ini File.
6. Reboot Constructor. You should now be able to see the Active Enterprise option at the top of the list of Business Processes in the Component Browser Area.

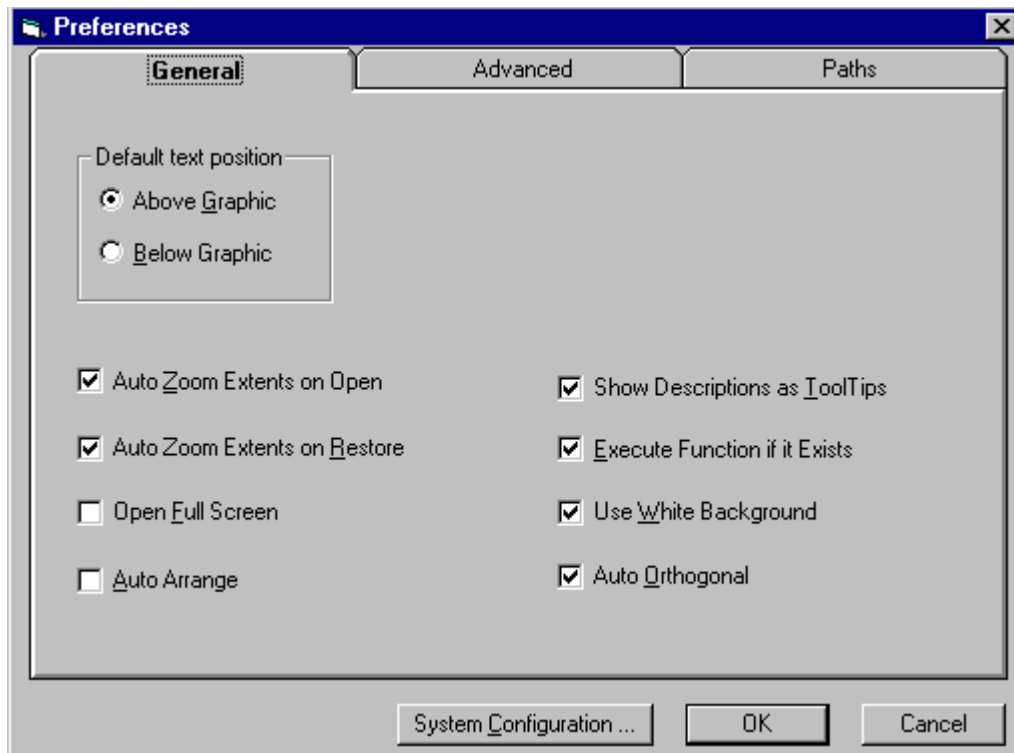


Designing Made Easier

To make designing easier we suggest that you select Preferences on the Edit Menu and check the boxes as shown below.



Tip: This is all explained in the Constructor help but you do not need to look this up now.

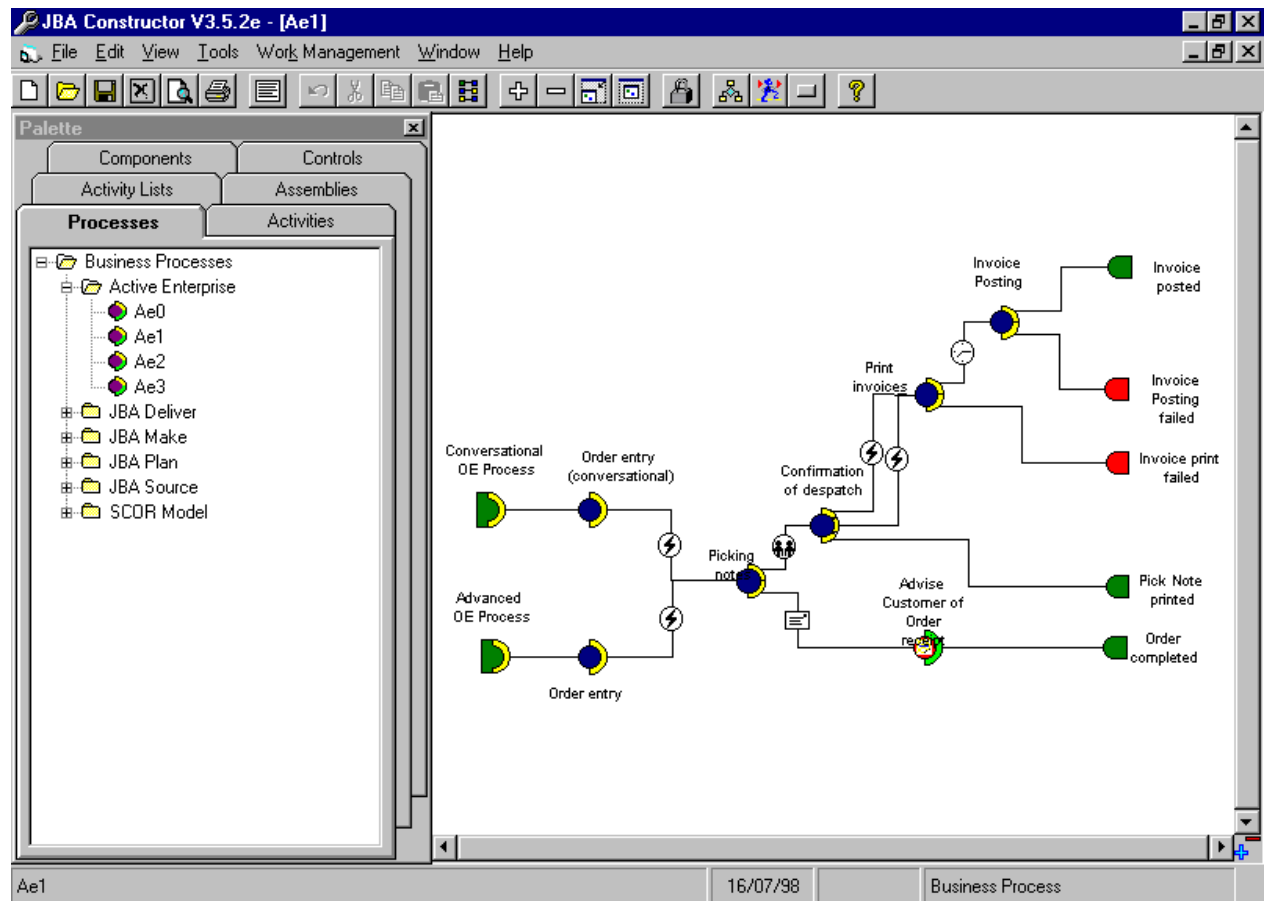


Note: The Open Full Screen checkbox enables you to toggle between showing the Component Browser Area or expanding the Workspace to fill the full screen. There is also a button on the toolbar which you can use for the same purpose. This button toggles between these two displays:



Building Ae1

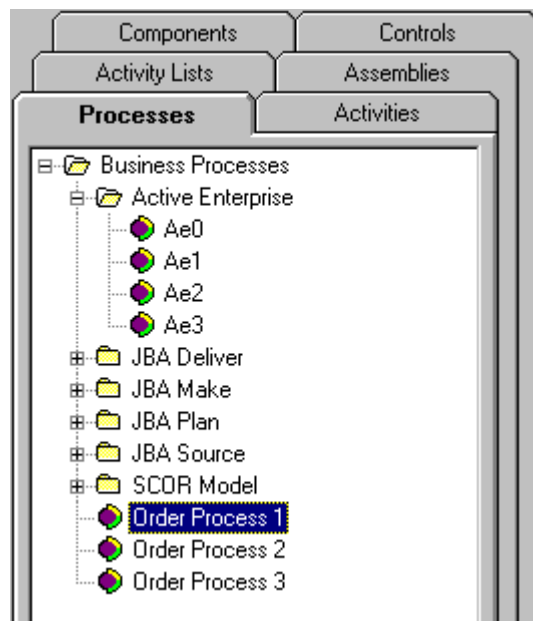
Ae1 is the name of the first Active Enterprise Business Process. Select Ae1 from the Component Browser Area on the left side of the screen, and double click the symbol to open it onto the workspace. It is in the Processes folder.



This tutorial shows you how to recreate this Business Process and save your copy with a new name. Yours will be called Order Process 1.

You can minimise Ae1 if you want to. You will not be making any changes to Ae1, Ae2 or Ae3.

At the end of this tutorial you will have created Order Process 1, Order Process 2 and Order Process 3. This is how you will see them in the Component Browser Area:



Note: We will not be telling you to click the OK button each time you finish entering details or to do a File and Save.

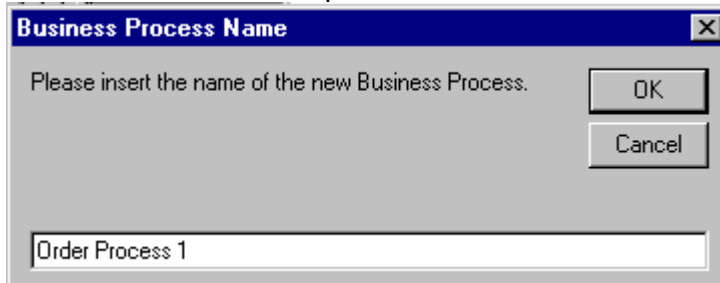
Saving Your Business Processes

When you design your Business Processes you will be saving them on the design-time database on your PC. You cannot view them using Windows Explorer but you can see them listed in the Component Browser Area.

It is only when you Activate the Business Processes, at the end of the tutorial, that they are then copied to the server.

Step 1 Start A New Process

1. First create a new name for the Business Process. Drop down the File Menu and select New and Business Process.
2. Enter a name. For example, Order Process 1.

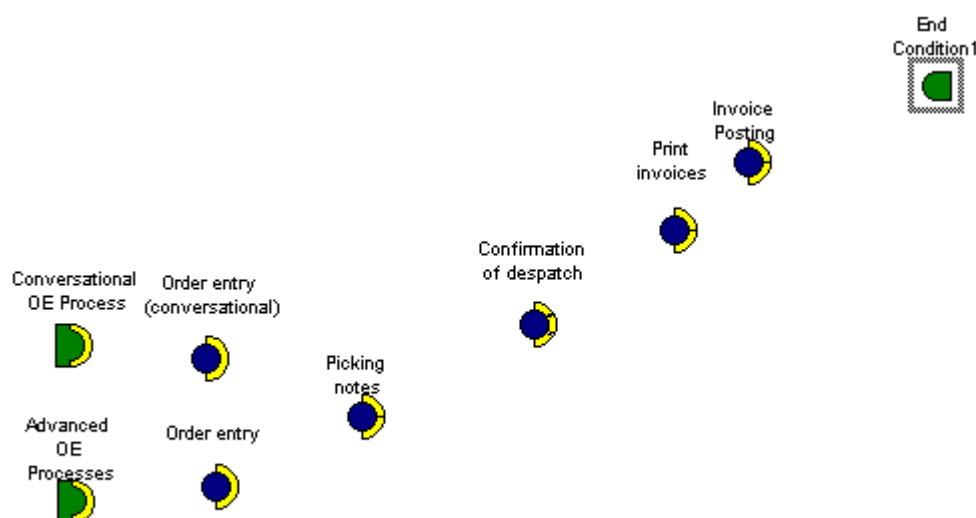


3. A new window is displayed for you on the Workspace with the new name in the title bar. Note that the name Order Process 1 now appears at the bottom of the Component Browser Area. Note that you need to save your work now, after each step.
4. Note that the Start Condition and End Condition are displayed for you.
5. Now change the name of the Start Condition to something more meaningful. To do this right click on the symbol and select Edit from the pop up menu.
6. Change the name to Conversational OE Process. Select Sales Order for the Document Type.
7. Note that you cannot enter the Process Code from here. That is because the Process Code applies to the whole Business Process and here you are looking at the details for just this Start Condition.
8. Now to enter the Process Code. For this you need the Edit Properties Window. Right Click on the Workspace and select Properties. Enter OP1 for the Process Code.
9. Now to enter another Start Condition. Again right click on the Workspace but select New and Start Condition.
10. This time change the name to Advanced OE Process. (Again select Edit from the pop up menu.) Note that OP1 is displayed for you now as the Process Code. Again select Sales Order for the Document Type.

Step 2 Add The Activities

Now you are going to select various activities from the Component Browser Area and drag them out onto the design for the Business Process. You have to locate each Activity in the lists in the Component Browser Area.

This is what you are aiming for:



Warning: Please follow the instructions below very closely. There can be more than one activity with the same name and it is important to open the correct one. If your activities do not have the same names or number of nodes as the ones in this Help text please check again. Under no condition be tempted to amend the Properties for these activities. The symbols for Elemental Activities reflect the exit conditions coded in the RPG program, for example, if the symbol has three exit nodes then this reflects the fact that the RPG program has three modes of exit. If you change the Properties then you update the Design time database only and when you try to execute the activities the programs will not run.

This is what you need to do:

1. Open the Activities folder in the Component Browser Area.
2. Start by selecting these folder and subfolders:
 - Select Work Management Enabled Activities
 - Then Customer Services And Logistics
 - And then Customer Service
 - And then Sales Order Processing

- And then finally Processing and Order Enquiries. Note that there are two subfolders you need in here. Order Processing and Reports And Batch Processing.

3. Look in Order Processing and drag out Order Entry(Conversational).



Warning: Please don't change the names or you will update the Design Time Database.

4. Look in Reports and Batch Processing and then Document Preparation and drag out Picking Notes.
5. Look in Order Processing and drag out Confirmation of Despatch. Check that this is the one with three nodes.
6. Look in Reports and Batch Processing and then Document Preparation and drag out Print Invoices.
7. Look in Reports and Batch Processing and then Batch Processing and drag out Invoice Posting.
8. Finally look in Customer Service again but this time take Advanced Order Entry, then Processing and drag out Order Entry. This is going to be the second version of Order Entry on your Workspace.

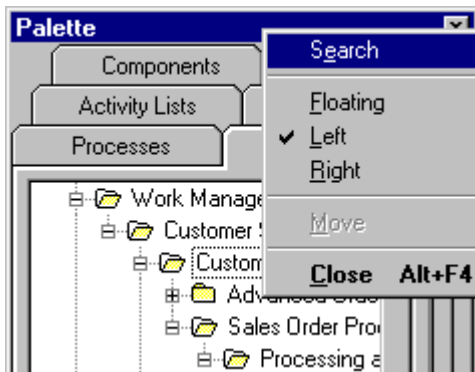
Step 3 Search For A Component



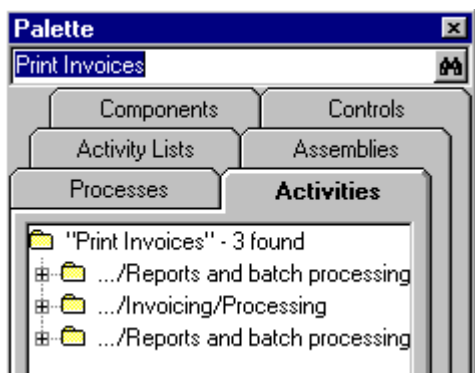
Tip: This is a useful facility so please try this now.

You can search for the name of any component in the Component Browser Area.

1. Move the cursor over the title bar in the Component Browser Area.
2. Right click and select Search



3. Enter the name for the search in the space provided.
4. Press return and the search results are displayed in place of the Component Browser.



5. To return to the Component Browser Area move the cursor over the title bar again right click and deselect Search.

Step 4 Resequence Nodes



Tip: This step prepares you for Step 5.

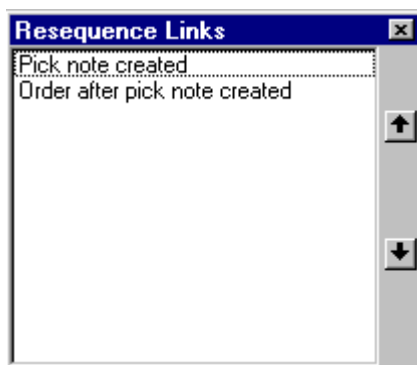
Most symbols have more than one node. These nodes represent the exit points from the component, for example, for OK and Cancel buttons. When you join up one symbol to another you have to make sure you have joined up the correct node.

For example, see Picking Notes. Hold the cursor over the two exit nodes to see the names:

- Pick Note Created. This goes on to join Confirmation Of Despatch.
- Order After Pick Note Created. This goes on to Order Completed.

You may find that the nodes are in the wrong order. You can reorganise the nodes easily and there are two methods:

- You can drag the nodes into a new position.
- You can select the Resequence Links option on the right click menu for the symbol. Select a node in the list and use the up and down arrows to reposition it.



Warning: You can reorder the nodes and it will affect only your design. However, don't edit the Properties or you will update the Design Time Database for all users.

Step 5 Add The Connecting Lines

You define a sequence for the Business Process by joining the components together with connection lines.

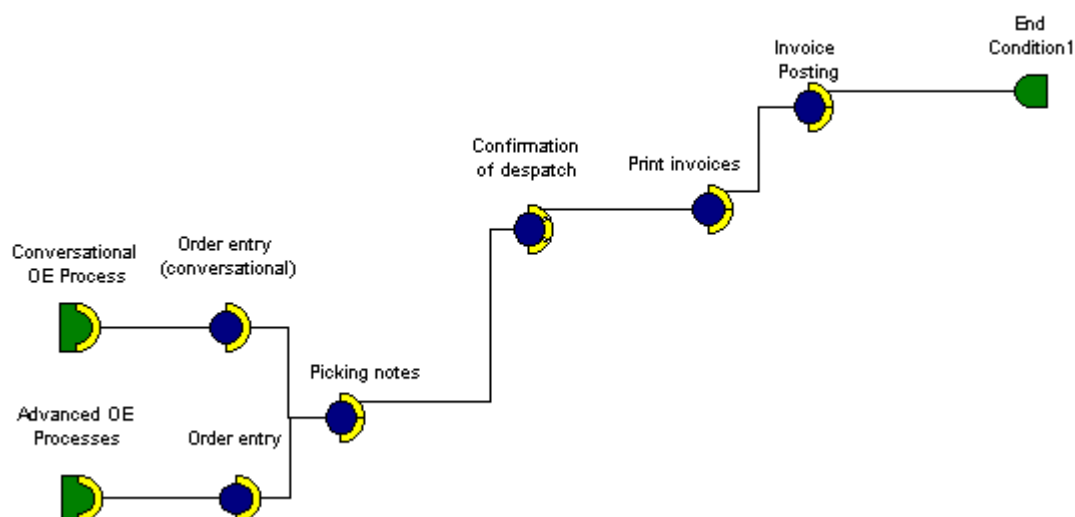
In order to create a connection line you click with the left mouse button on a node of one component, continue to hold the button down, and then release it when the cursor is over the circular part of the target component.

Join the symbols with connecting lines in the order in which you created them.



Tip: Check each node is the top one for each symbol. Refer to Step 4 if you need to resequence links.

This is what you are aiming for:



For Picking Notes select the Pick Note Created node.

For Confirmation Of Despatch select both the Consolidated Invoice Created/Updated node and the Invoice Generated node and join them both to Print Invoices.

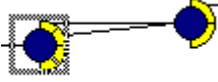
For Print Invoices select the Invoice Printed node.

For Invoice Posting select the Invoice Posted node.

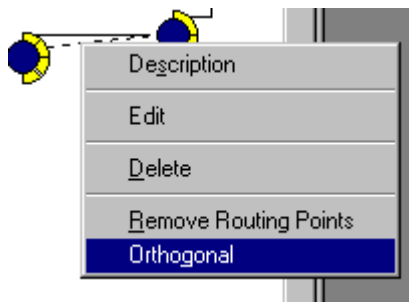
Design Tips

☺ Tip: You need to experiment to find the best positions for the symbols. Try the Arrange button on the Toolbar. This tidies your design.

Sometimes you get an untidy diagonal line.



You can right click on the diagonal line and select Orthogonal from the pop up menu and this puts right angles in for you.

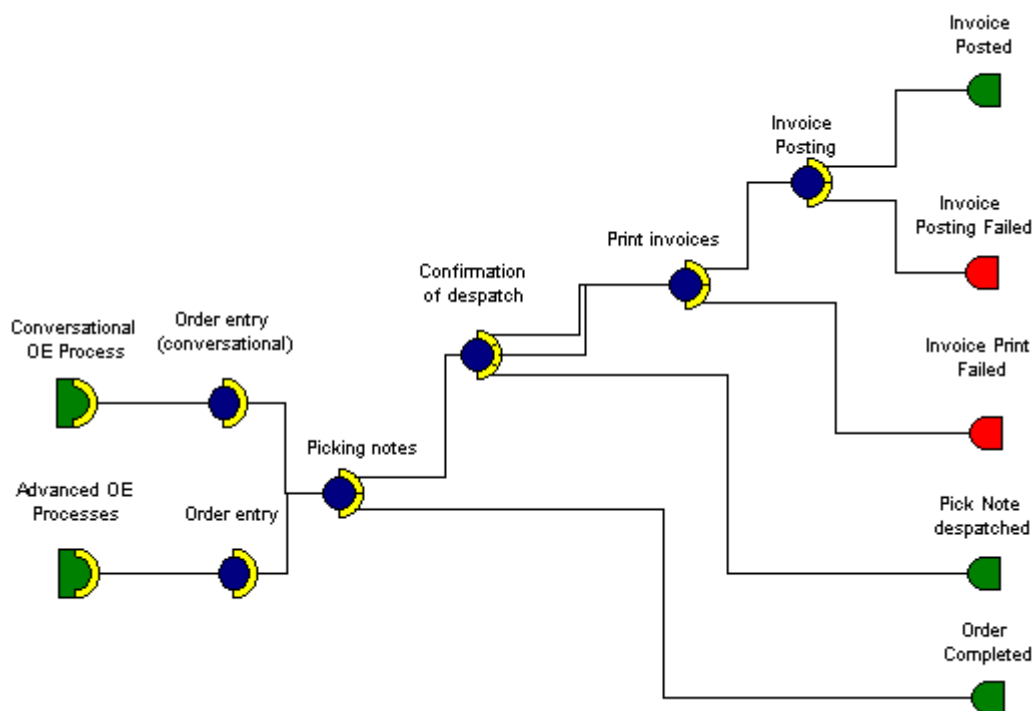


☺ Tip: You can set all lines to be automatically orthogonal. To do this select Preferences from the Edit Menu and check Auto Orthogonal on the General panel.

Step 6 Put In The End Conditions

You now need to end each pathway to complete the design. This design has five end conditions.

This is what you are aiming for:



There are two ways to create an End Condition: either drag out an End Condition from the Process Controls folder or right click on a blank space on the Workspace and select End Condition from the pop up menu.

Then to change the Name you select the End Condition and select Edit from the right click pop up menu.

1. For the existing End Condition change the Name from End Condition 1 to Invoice Posted.
2. Now put in four other End Conditions for:
 - Invoice Posted
 - Invoice Posting failed
 - Invoice Print failed
 - Pick Note printed.
3. Join the End Conditions to the symbols, making sure the nodes are correct. Hold the cursor over each node to display the tooltip.

Colours

There are two End Conditions which represent failures:

- Invoice Posting failed
- Invoice Print failed.

It is helpful to change the colour of these two to red to make them stand out. To do this you select the Colour/Select Colour option from the right click menu and choose the red colour.

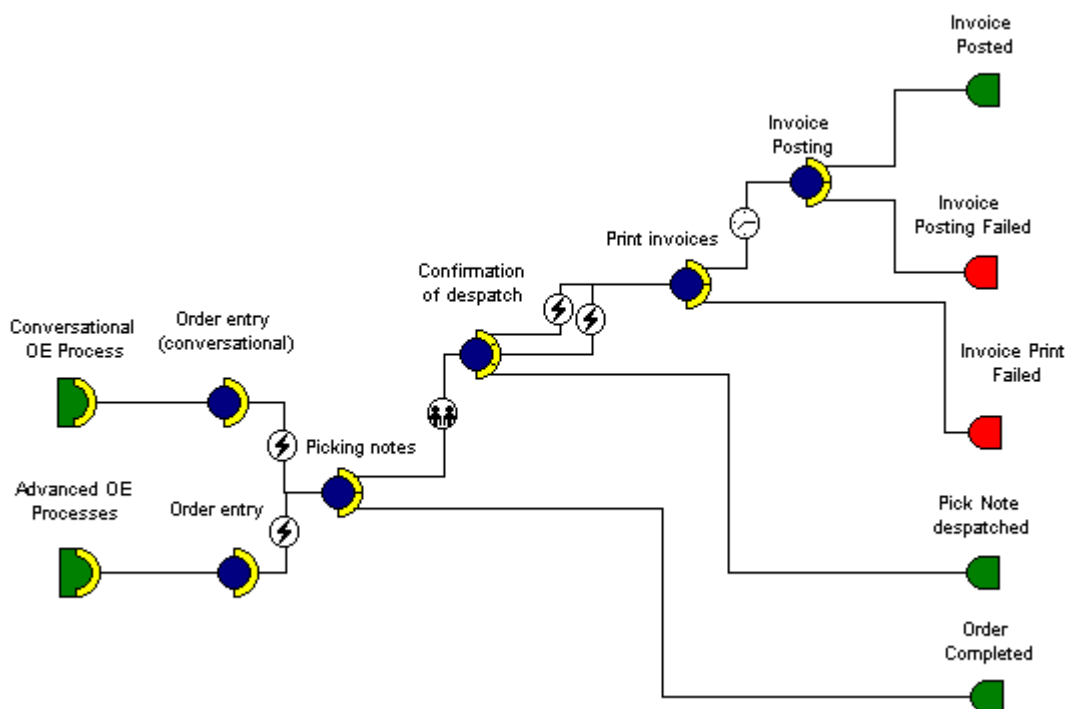
Step 7 Add Scheduling Information

Constructor lets you determine how you want each activity to be scheduled. You do this by placing symbols on the connection lines to indicate your choice of scheduling option. It is the connection line which holds the Scheduling Information.



Note: To delete a schedule you select the connection line and select Delete from the right click pop up menu.

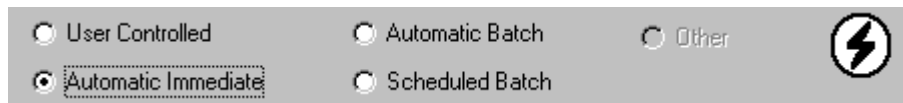
This is what you are aiming for:



Part 1 Automatic Immediate

You will be adding Automatic Immediate symbols.

1. Right click on the connection line going into Picking Notes from Order Entry(Conversational) and select Edit. This displays the Execution Details Window. Note that although you have two Order Entry symbols leading into Picking Notes, there are in fact two connection lines. If you highlight one of them you will see that the line is displayed as broken and you can distinguish between the two.
2. Note that there is an option button. Choose Automatic Immediate. This means the execution of the process will continue automatically.

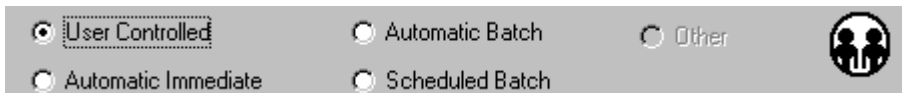


3. Similarly add an Automatic Immediate symbol between Order Entry and Picking Notes.
4. Similarly add an Automatic Immediate symbol for both pathways between Confirmation of Despatch and Invoice Posting.
5. Select File/Save.

Part 2 User Controlled

Add a User Controlled symbol between Picking Notes and Confirmation of Despatch.

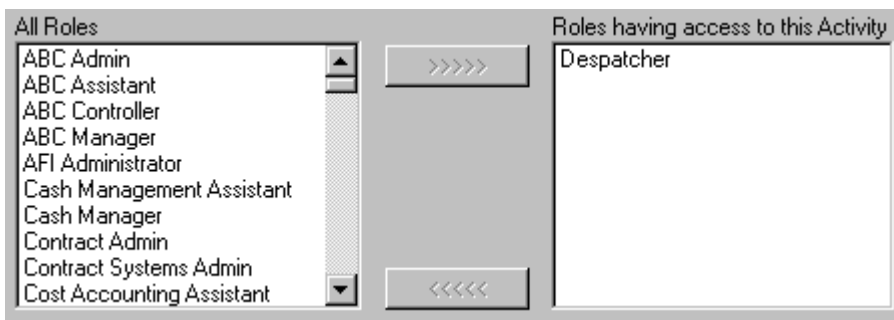
1. Right click on the connection line going into Confirmation of Despatch and select Edit. This displays the Execution Details Window.
2. Note that there is an option button. Choose User Controlled.



User Controlled means that the user intervenes in the execution of the process by making a decision about which pathway the processing will follow. For example, by choosing whether to contact a customer about a failed credit check or cancel the order.

For this you also have to choose which role user is to have control of this part of the process.

3. Select the Role panel and check that you have Despatcher somewhere in the right hand list. If not then select the Despatcher from the left hand list and click the right arrows button to move it across.



4. Select the Action List panel and select Despatcher for the recipient and click OK.



This means that when this process is executed the Despatchers will see this activity assigned to them via their Action Lists.

5. Select File/Save.

Part 3 Scheduled Batch

Add a Scheduled Batch symbol between Print Invoices and Invoice Posting.

For Interest Only

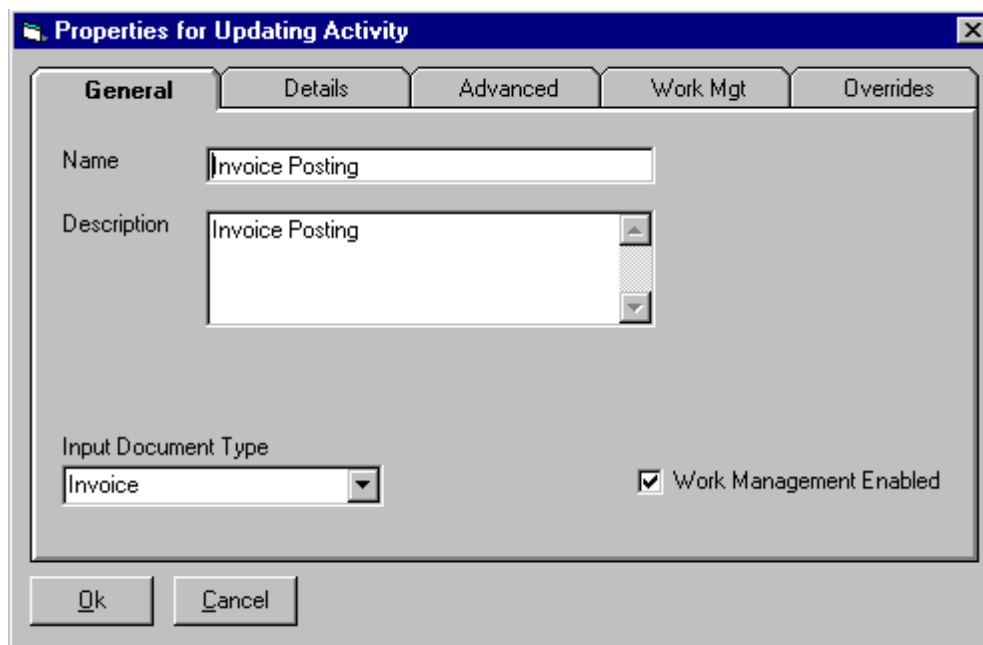
Look at the Properties for the Invoice Posting activity. This will help you to understand the scheduling process.



Warning: Do not edit Properties - you will update the Design Time database for everyone.

This is what you do:

1. Select Properties on the right click menu for Invoice Posting.
2. Note that the Work Management Enabled box is checked on the Properties General panel



3. Select the Work Mgt panel.

The screenshot shows the 'Properties for Updating Activity' dialog box with the 'Work Mgt' tab selected. The 'Global Activity' checkbox is unchecked. The 'Activity Code' field contains 'PSTINV' and the 'Error Role' dropdown is set to 'System Admin'. Below these is a table titled 'Execution modes'.

Mode	Description	Processor	Processor type	Function
30	Automatic Batch	OE 033180	TASK	
40	Automatic Scheduled	OE 033180	TASK	

At the bottom of the dialog are 'Ok' and 'Cancel' buttons.

5. Note that there is a unique Activity Code - PSTINV.
6. Note that there is a 40 entry in the grid box for the Mode.
7. Click Cancel.

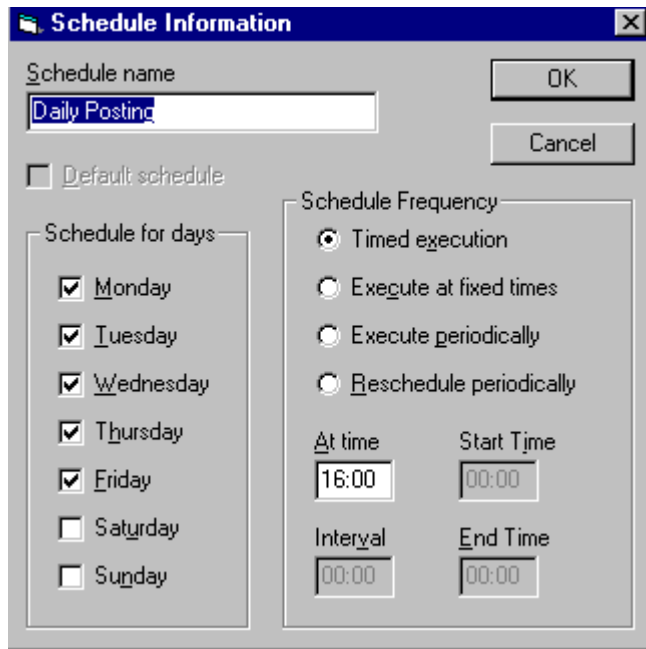
Entering Scheduling Details

1. Right click on the connection line going into Invoice Posting and select Edit. This displays the Execution Details Window.
2. Note that there is an option button. Choose Scheduled Batch.

This screenshot shows four radio button options for scheduling: 'User Controlled', 'Automatic Batch', 'Other', and 'Scheduled Batch'. The 'Scheduled Batch' option is selected. To the right of these options is a clock icon.

Scheduled Batch means the execution of the process will continue as part of a queue of batch jobs scheduled to start at a certain time.

3. You are already in the Schedule Panel. Click on the Change button to create a schedule. This displays the Select Schedule Window.
4. Click on the New button. This displays the Schedule Information Window.
5. Enter Daily Posting for the Schedule Name, select Timed Execution and enter 16:00 for the At Time as follows.



Schedule Information

Schedule name:

☐ Default schedule

Schedule for days:

- ☒ Monday
- ☒ Tuesday
- ☒ Wednesday
- ☒ Thursday
- ☒ Friday
- ☐ Saturday
- ☐ Sunday

Schedule Frequency:

☒ Timed execution

☐ Execute at fixed times

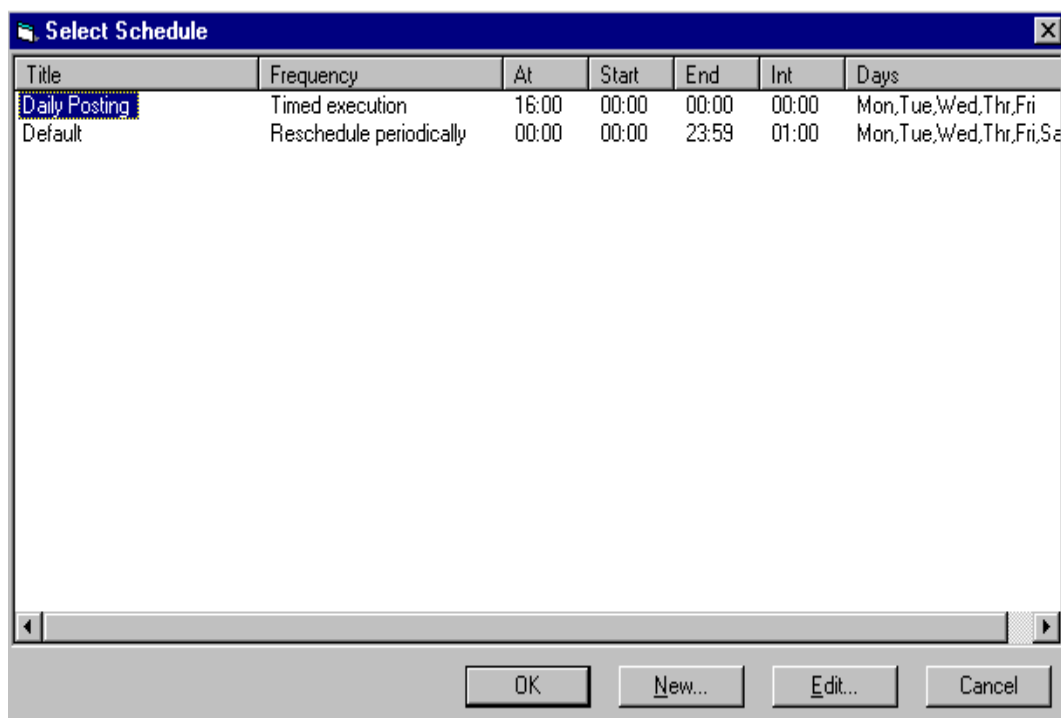
☐ Execute periodically

☐ Reschedule periodically

At time: Start Time:

Interval: End Time:

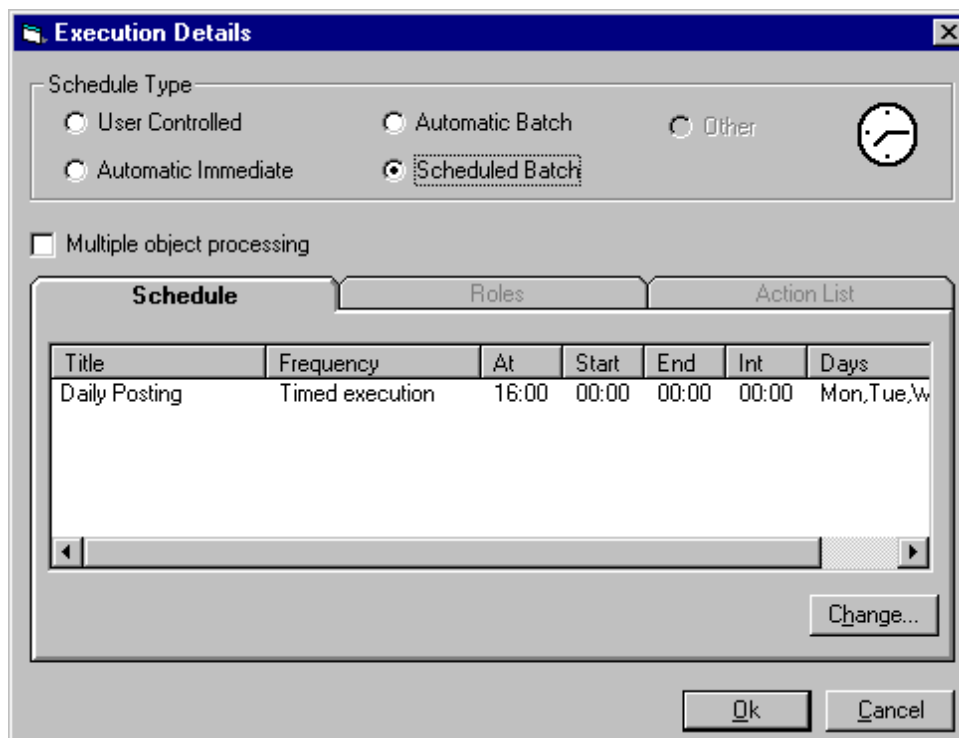
6. Click OK to return to the Select Schedule Window, displaying your new schedule in the list.



Select Schedule

Title	Frequency	At	Start	End	Int	Days
Daily Posting	Timed execution	16:00	00:00	00:00	00:00	Mon,Tue,Wed,Thr,Fri
Default	Reschedule periodically	00:00	00:00	23:59	01:00	Mon,Tue,Wed,Thr,Fri,Sa

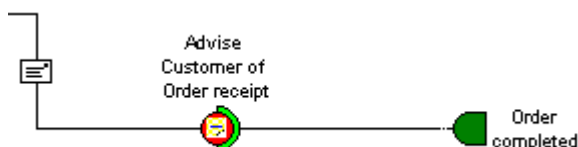
7. Make sure that the new schedule is selected and click OK.
8. Check that the Execution Details Window now looks like this below and you can click OK. If it looks different then click the Change button and change the details again. It may be that you did not have the correct schedule selected before you clicked OK.



9 Note that you haven't used the Schedule Types called Automatic Batch and Other. Automatic Batch means the execution of the process will continue automatically but will form part of a queue of batch jobs. The Other option is available only if you have defined your own execution modes. You don't need to worry about these for now. If you do find that you need them at some time then please refer to the online help available with Constructor, where all this is fully documented within the topic called Execution Details Window.

10. Please select File/Save.

11 If you compare your design with Ae1 you will see that the connection line leading to Order Completed has two symbols missing as shown below. This is an Email symbol and a Manual Activity symbol.



You will learn how to create Manual Activities in the Building Ae2 section of the tutorial and Email will be covered in the Building Ae3 section. You will be reminded then to come back to Ae1 and add in these symbols. In every other respect the Business Process Ae1 is now complete. Check again that you have saved your design.

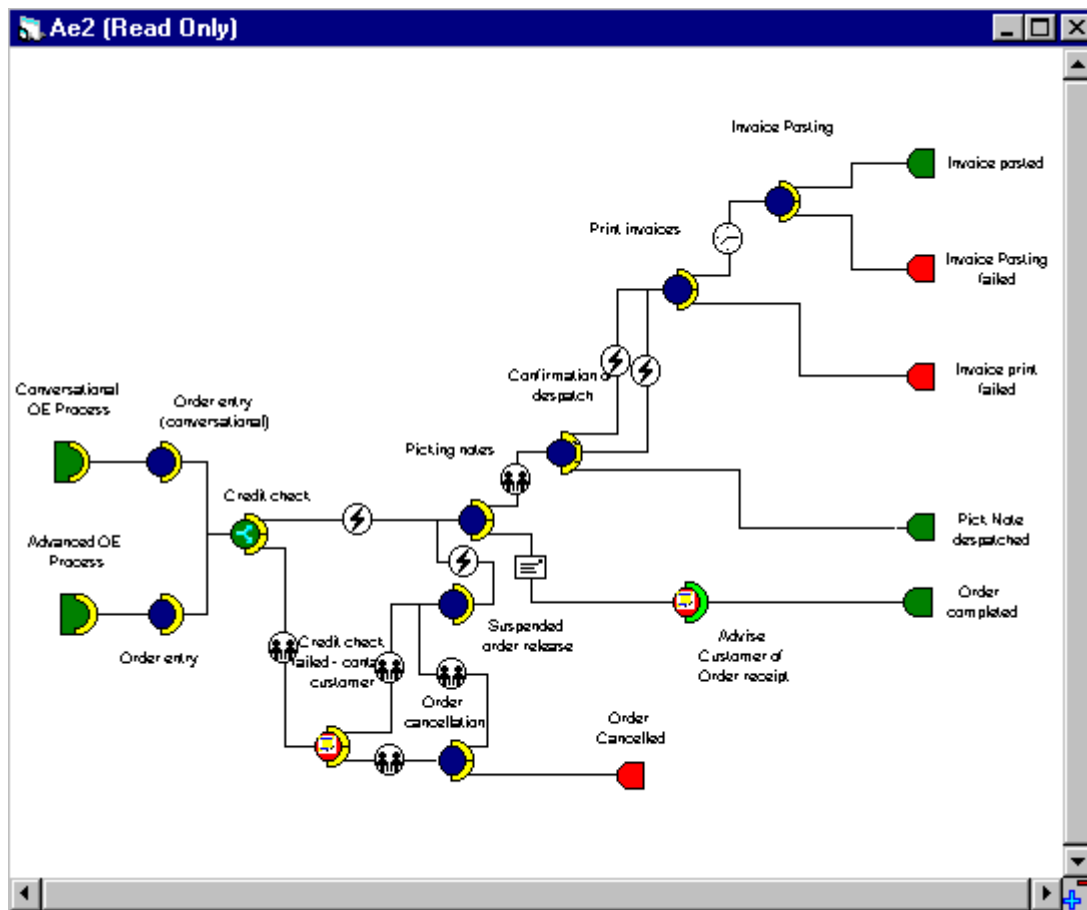
12 Note that you could Validate and Walkthrough the design at this stage. This hasn't been explained yet and for the purposes of this tutorial we recommend that you do this once at the end, after building Ae3. It will be explained fully then.

14 Now please refer to the help text for Building Ae2.

Building Ae2

 **Warning: Ae2 builds onto Ae1. Please make sure you follow the tutorial for Ae1 first.**

Ae2 is the name of the second Active Enterprise Business Process. Select Ae2 from the Component Browser Area on the left side of the screen and double click the symbol to open it onto the workspace.



This tutorial shows you how to recreate this Business Process and save your copy with a new name. Yours will be called Order Process 2.

You will be adding extra functionality to handle credit checking failure.

 **Tip:** Look at Ae2 and note the new symbols between and below Order Entry and Picking Notes.

Step 1 Create A New File

1. Select the Window option on the Main Menu and select the Order Process 1 option.



Tip: If you haven't done this previously select File/Save to make sure you save Order Process 1.

2. Select File/SaveAs and change the name to Order Process 2.
3. Right click on the Workspace, click Properties and change the Process Code to OP2 and save again.
4. A second window is displayed for you on the Workspace with the new name in the title bar. Minimise Order Process 1, Ae1 and Ae2 if you want to.
5. You may need to drag Start Conditions and Order Entry symbols across the Workspace to the left to make more room to the left of Picking Notes.
6. Open the Activities folder on the Component Browser Area.
7. Look in:
 - Work Management Enabled Activities
 - Customer Services And Logistics
 - Customer Service
 - Sales Order Processing
 - Processing And Order Enquiries
 - Order Processing.
7. Drag across Order Cancellation and Suspended Order Release.

Step 2 Add An Action Agent



Note: An Action Agent is a hidden piece of programming. In this example you tell the Business Process how to handle credit checking after the processing for Order Entry. The Document Type is Sales Order coming out of Order Entry.

Now you are going to create a Credit Check Action Agent between Order Entry and Picking Notes.

1. Right Click and select New Action Agent from the pop up menu.
2. This displays the Action Agent Path Window.

3. Change the name to Credit Check and change the Description to: 'Server executed Action Agent to check the percentage of the available credit limit that a customer has used'. Click OK.
4. Join up Order Entry to Credit Check.

The easiest way to do this is to delete the lines which you already have between the Order Entry symbols and Picking Notes. Note that this also deletes the Automatic Immediate symbols. Position the Action Agent alongside and to the left of Picking Notes. Now you can join up Order Entry to Credit Check Action Agent. Then join up Credit Check Action Agent to Picking Notes and re-create the Automatic Immediate symbol leading into Picking Notes.
5. Now select Properties from the Credit Check Action Agent right click menu. Because you have joined up the Action Agent the window expands now to show:
 - a) That the Document Type coming out of Order Entry is Sales Order
 - b) The exit nodes from the Action Agent. Note that only the default node is showing. Your Action Agent currently has only one exit node but you need to give it another one.
6. You need to define what the process needs to do in order to handle exceptions from the Order Entry activity. You set a test for the Credit Usage and if it fails you can direct the processing into credit failure processes.

Enter details as shown below in order to test whether the customer's credit has exceeded 100. To do this place the cursor over the word (Default) and right click and select Add. This

adds a second row into the grid box. Select each cell in the row and you will be able to select options from a drop down list or enter text.

This creates a new node on the Action Agent symbol. The default, where the credit usage has not exceeded 100, is to continue on to process Picking Notes.



Warning: If Customer Credit Usage Percentage does not appear in the browse list for the Input Condition field do not attempt to create it. Consult the person who manages your database.

Properties for Action Agent

Name: Document type:

Description:

Title overrides:

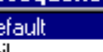
Define exit node rules

Code	Combination	Input Condition	Operator	Test Condition	Display Name
01	1st Rule	Customer Credit Usage Percentage	>	100	Fail
(Default)					Default

☐ Save definition to palette on exit

OK Cancel

7. Check the nodes of the Action Agent on the Workspace. Default should be at the top and Fail at the bottom. If they are the wrong way round resequence them using the Resequence Links option on the right click menu.



8. Join up Default to Picking Notes. Check that you have an Automatic Immediate schedule on the connecting line.
9. Remember to join up Fail to the manual activity which you create next.

Step 3 Add Manual Activities



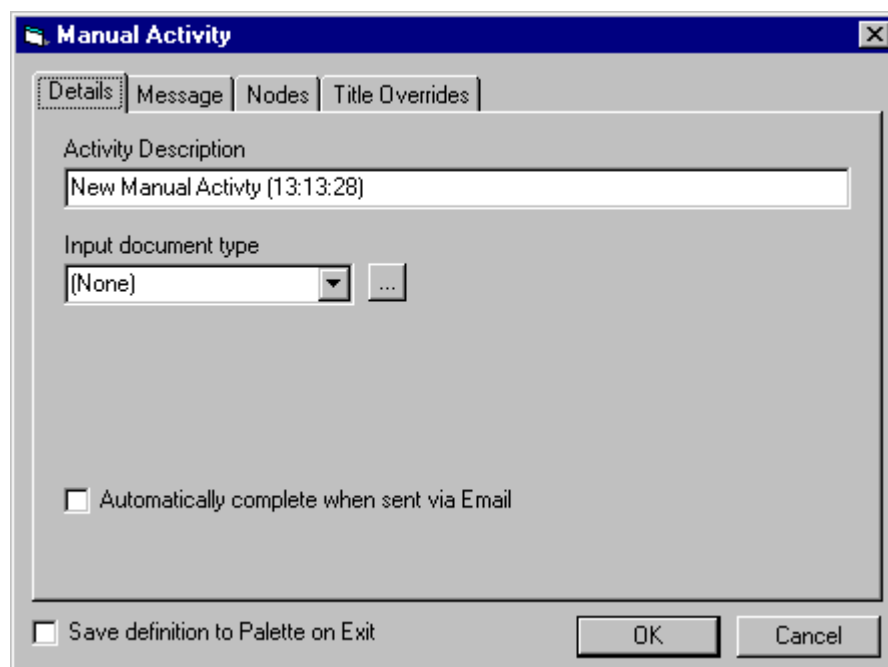
Note: A Manual Activity is a way of allowing the user to have an input to the process. In this example the user will be able to decide what to do about failed credit checks. The Document Type is still Sales Order coming out of Order Entry.

Now you are going to create a Manual Activity following on from the Fail node from Credit Check Action Agent.

Right Click and select first New and then Manual Activity from the pop up menu.

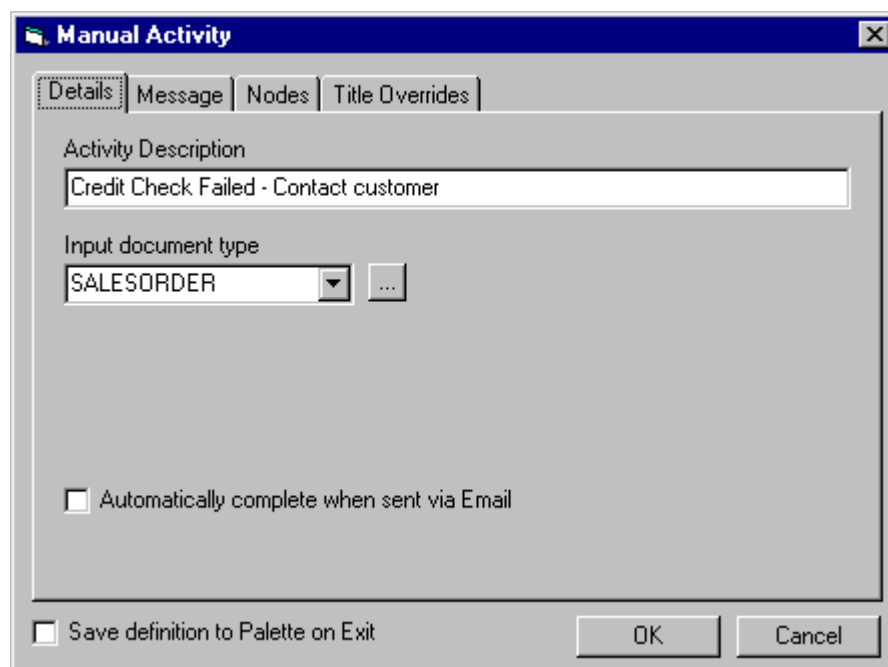
The Manual Activity Window is displayed. There are four panels which you must complete as follows.

Details Panel



The 'Manual Activity' dialog box is shown with the 'Details' tab selected. The 'Activity Description' text field contains 'New Manual Activity (13:13:28)'. The 'Input document type' dropdown menu is set to '(None)'. There is a checkbox for 'Automatically complete when sent via Email' which is unchecked. At the bottom, there is a checkbox for 'Save definition to Palette on Exit' which is unchecked, and 'OK' and 'Cancel' buttons.

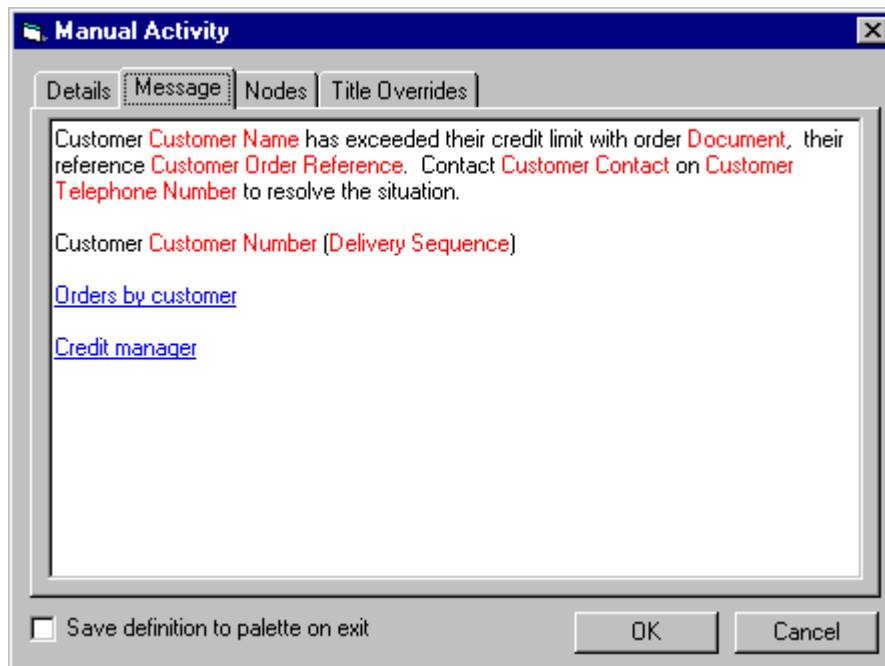
Change the Activity Description and select the Input Document Type as follows.



The 'Manual Activity' dialog box is shown with the 'Details' tab selected. The 'Activity Description' text field now contains 'Credit Check Failed - Contact customer'. The 'Input document type' dropdown menu is now set to 'SALESORDER'. The checkbox for 'Automatically complete when sent via Email' remains unchecked. At the bottom, the checkbox for 'Save definition to Palette on Exit' remains unchecked, and the 'OK' and 'Cancel' buttons are present.

Message Panel

You are now going to complete the Message panel details as shown below.



First please note that this determines what the user will see on the screen. You can enter three kinds of text as follows:

Black

The black text is actual text which the user sees, for example, '...has exceeded their credit limit...'

Red

The red text is Variable Data, for example, Customer Name. To insert red text fields use the right click menu and select Insert Data Field.



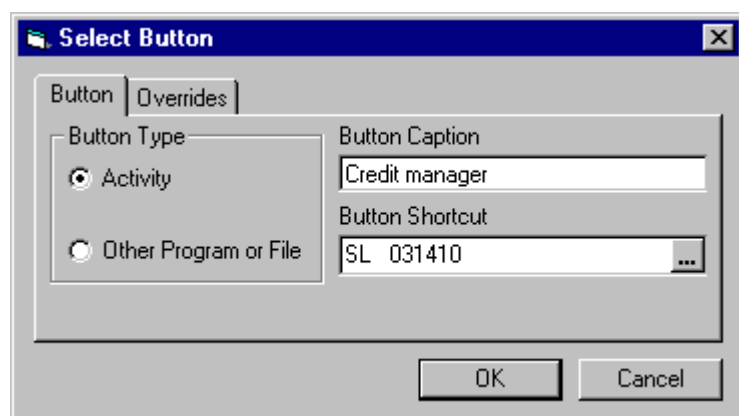
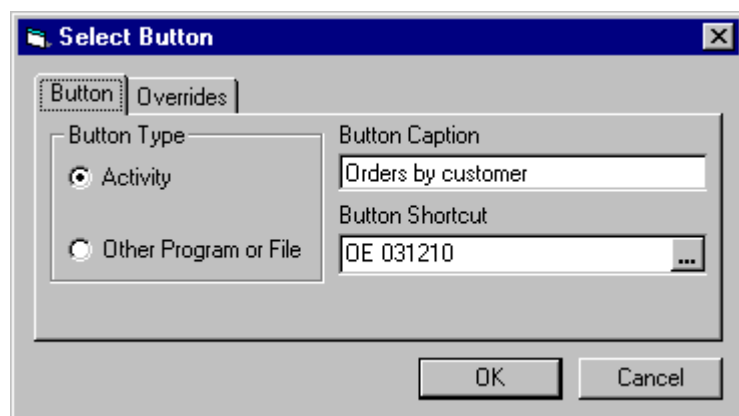
Warning: If a data field does not appear in the browse list do not attempt to create it. Consult the person who manages your database.

Blue

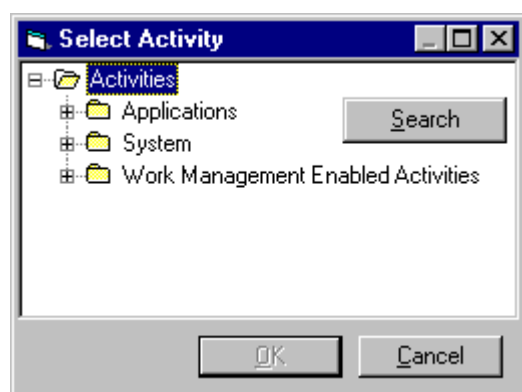
The blue underlined text represents a button on the screen which the user can see and use. For example, Orders by customer. These buttons represent pathways which go off at a tangent to the processing and do not interfere in any way with the logical flow through the Business Process.

For this example you need to add two buttons. You do this by selecting Insert Button on the right click menu. This displays the Select Button Window.

If you know the shortcut code you can enter the details directly as follows.

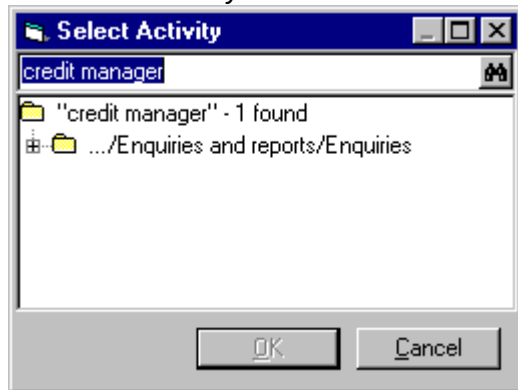


However, for practice we suggest that you try the search method now. Please ignore the Button Caption field and click the three-dots browse button alongside the Button Shortcut field. This displays the Select Activity window. Right click on the window to display a Search option on a pop up menu as follows.



Click Search to display a text box at the top of the window. Enter search details, for example, Credit Manager and press Return. This starts the search.

The search tells you if a match has been found and lists the results in a browser menu format.



Warning: If your search finds no match then please report this to the person who manages your database.

Select the activity from the list of items you have just found and click OK. This returns you to the Select Button Window and completes all the details you need.

Note that the Message text also appears on the tooltips for the activities.

Nodes Panel

1. Select the Nodes Panel.
2. Complete the nodes details as below. To make the design more meaningful change OK to Continue and change Cancel to Cancel Order.

The screenshot shows a dialog box titled "Manual Activity" with a close button (X) in the top right corner. It has four tabs: "Details", "Message", "Nodes", and "Title Overrides". The "Nodes" tab is selected. Inside the tab is a table with three columns: "Node", "Caption", and "Completion Action". The table has two rows of data. The first row has "02" in the "Node" column, "Cancel Order" in the "Caption" column, and an empty "Completion Action" cell. The second row has "01" in the "Node" column, "Continue" in the "Caption" column, and an empty "Completion Action" cell. Below the table is a checkbox labeled "Save definition to palette on exit". At the bottom right are "OK" and "Cancel" buttons.

Node	Caption	Completion Action
02	Cancel Order	
01	Continue	

3. Check on the Workspace that these nodes are in the correct sequence. You may need to select Resequence Links on the right click menu and reverse the order so that Continue is the node at the top of the symbol and Cancel Order is at the bottom.

What you have done is to create a set of choices for the Explorer users, so that when they run the Credit Check activity, and the credit check fails, they will see a screen called Credit Check failed - Contact Customer. This screen will have five buttons as follows:

- Defer button. This is always displayed for the user, as part of the Manual Activity process. This enables the user to leave the Manual Activity on the Action List, halted at this point, ready for later processing.

When they click one of the following buttons they determine the next pathway through the Business Process:

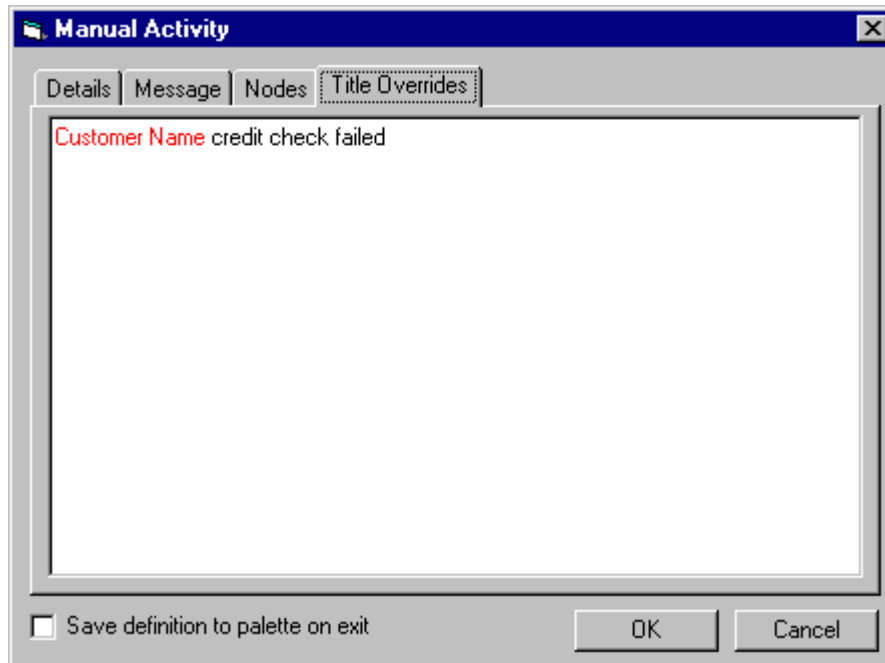
- Continue button, which goes off to Suspended Order Release (as shown on the Workspace)
- Cancel Order button, which goes off to Order Cancellation (as shown on the Workspace)

The following two buttons will be available without affecting the process and are there so that the user has the tools initially to resolve the activity and decide which route to take next.

- Credit Manager button, which goes off to the Credit Manager program (at a tangent to the processing on the Workspace)
- Orders By Customer button, which goes off to the Orders By Customer program (at a tangent to the processing on the Workspace).

Title Overrides Panel

Select the Title Overrides panel. The text you enter here will override the title of the Manual Activity when the user sees the screen. The user in this example would see 'Customer Name credit check failed', where Customer Name is a variable displaying actual text. For example, the user could see ' A.E.Smith credit check failed'.



Tip: The Save definition to Palette on Exit field on this window is very useful for copying Manual Activities from one design to another. You can copy to the clipboard and the name of the activity appears in the list of activities in the Component Browser Area. Then you can drag it out to another Business Process on the Workspace.

Click OK to return to the Workspace.

This is what is left to do regarding this Manual Activity:

1. Check that you have joined up the Credit Check Action Agent to this Manual Activity.
2. Join up the Continue node to Suspended Order Release.
3. Join up the Cancel Order node to Order Cancellation.

Complete Ae2

Please work through the list below, in strict sequence, to complete the design for Order Process 2. Everything you need to do has been covered already in the tutorial:

1. Join up the Order Lines Cancelled node, on the Order Cancellation activity, to Suspended Order Release. Check the node names as you do this.
2. Put a User Controlled schedule symbol before all the new activities you have created for Order Process 2. Remember that you select each connecting line and use Edit on the right click menu. You will need to select a Recipient Field on the Action List panel. You use the Roles panel to make the roles available for selection in the Recipient Field. These are the Recipients to use:
 - For the Fail node connection line from the Credit Check symbol please use Sales Admin.
 - For the connection lines between Credit Check Failed, Order Cancellation and Suspended Order Release please use Order Entry Clerk..
3. Add an End Condition, give it the Name of Order Cancelled and link to Sales Order Cancelled node in Order Cancellation Activity.
4. Link Suspended Order Release activity to Picking Notes with an Automatic Immediate schedule on the link.
5. Put in a Manual Activity between Picking Notes and Order Completed. Please complete the panels as shown below. Note that the Nodes panel correctly shows the OK node only.

Manual Activity

Details | Message | Nodes | Title Overrides

Activity Description
Advise Customer of Order Receipt

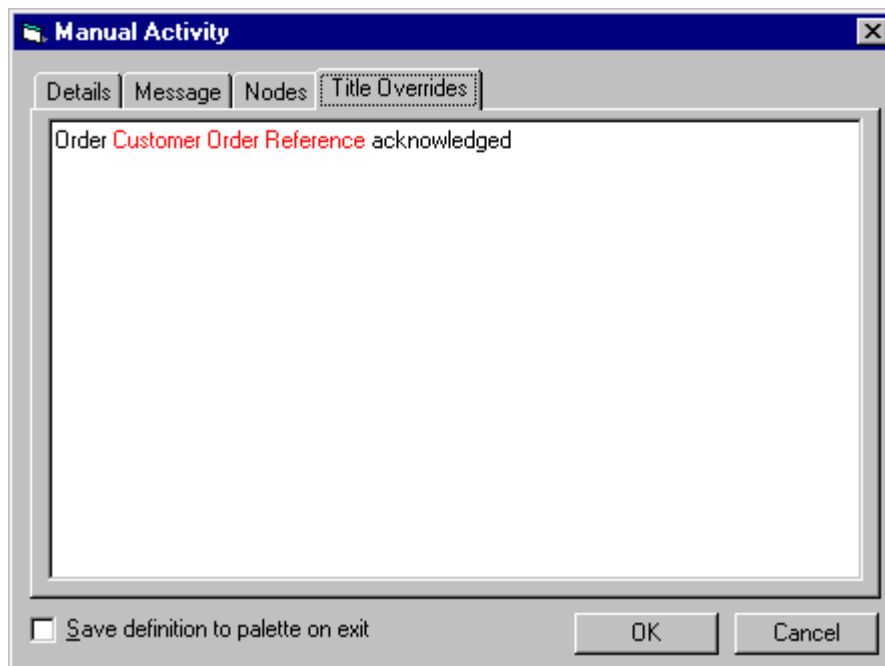
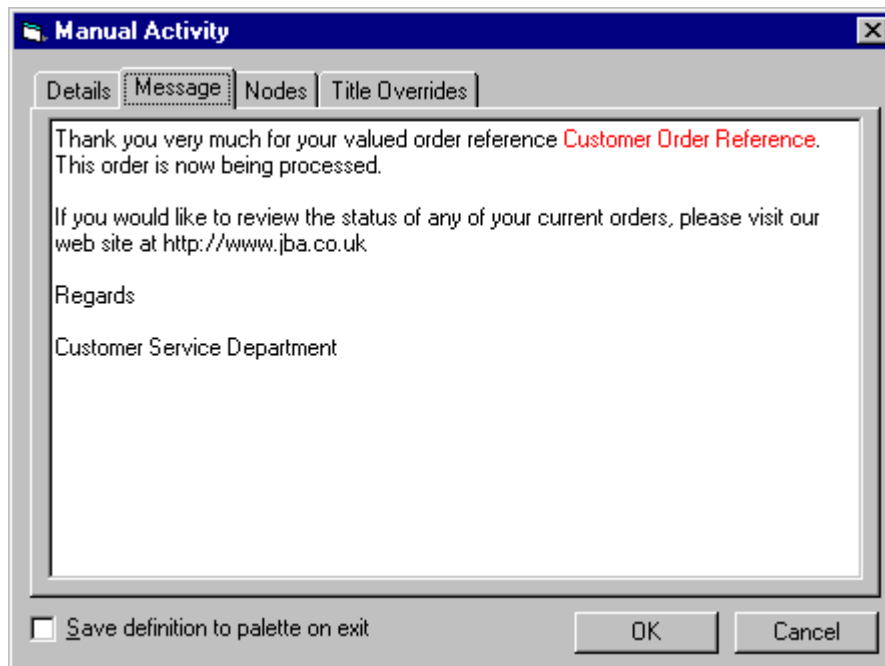
Input document type
Sales Order

☒ Automatically complete when sent via Email

☐ Save definition to palette on exit

OK Cancel

Please note the checkbox called Automatically Complete When Sent Via Email. You must check this box if you are sending Manual Activities by Email. Otherwise the Action Tracker will permanently report this as Pending, with a red icon, even when the Email has been sent.



Warning: Now open your design for Order Process 1 and put in the same Manual Activity again. This is something that you missed out earlier. Remember to save Order Process 1.

6. Check your design for Order Process 2 against Ae2 and you should notice that Ae2 has an Email symbol before Advise Customer Of Order Receipt. We recommend that you leave this for now and do it in Ae3 where this schedule is explained in detail. You will be reminded again to come back to Order Process 1 and Order Process 2 and add in the Email symbol.
7. The Order Process 2 is now complete for the time being. Remember to save it. Please continue now with the help text for Building Ae3.



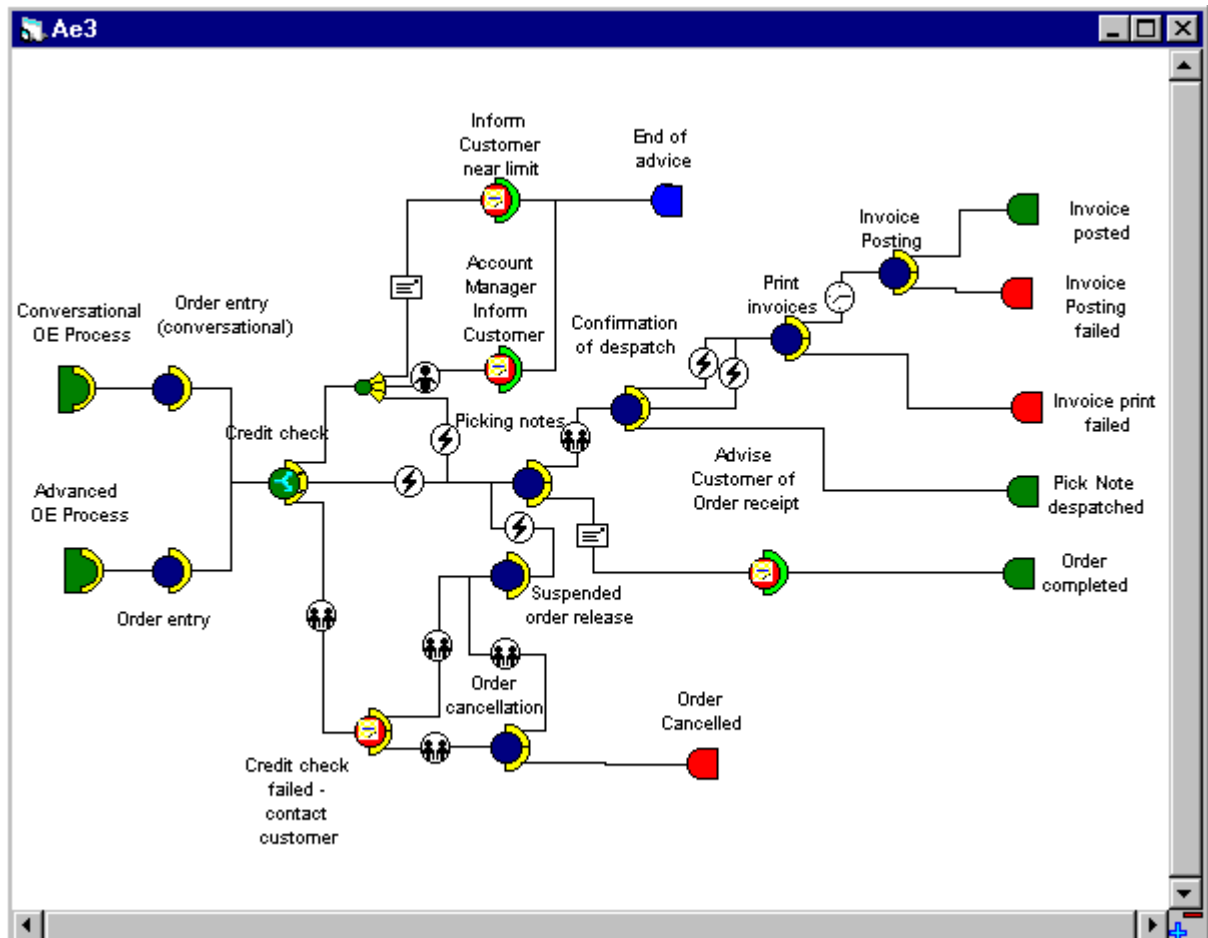
Note: You could Validate and Walkthrough the

design at this stage. However, for the purposes of this tutorial we recommend that you do this once at the end, after building Ae3. It will be explained then.

Building Ae3

Warning: Ae3 builds onto Ae2 and Ae2 builds onto Ae1. Please make sure you follow the tutorials for Ae1 and Ae2 first.

Ae3 is the name of the second Active Enterprise Business Process. Select Ae3 from the Component Browser Area on the left side of the screen and double click the symbol to open it onto the workspace.



This tutorial shows you how to recreate this Business Process and save your copy with a new name. Yours will be called Order Process 3.

You will be adding extra functionality to handle credit checking for notification when limits are nearly reached.

Tip: Look at Ae3 and note the new symbols above and between Credit Check and Picking Notes.

Step 1 Create A New File

1. Select File/Save to make sure you save Order Process 2.
2. Select File/SaveAs and change the name to Order Process 3.
3. Right click on the Workspace, click Properties and change the Process Code to OP3 and save again.
4. A third window is displayed for you on the Workspace with the new name, Order Process 3, in the title bar. Minimise any other windows if you want to.

Step 2 Add A Third Credit Checking Node



Tip: First zoom in on the Credit Check Action Agent. Use the plus and minus signs at the bottom right of the screen and click with the left mouse button.

Remember that in Ae1 there was no credit checking. Then for Ae2 you added this Action Agent so that you could put in new processing to handle credit checking failure. This is how the Credit Check Action Agent looks at present with two nodes:

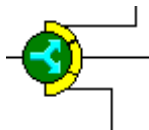


Default - successful credit check goes on to Picking Notes.

Fail - unsuccessful credit check goes on to the Manual Activity called Credit Check Failed - Contact Customer

Now for Ae3 you are going to put in new processing to inform customers who are near their limit. To do this you are going to add a third node called Inform.

The nodes will look like this:



Inform - successful credit check but customer is near credit limit and may be informed.

Passed(default) - successful credit check goes on to Picking Notes.

Fail - unsuccessful credit check goes on to either Order Cancellation or Suspended Order Release

This is what to do:

1. Select Properties from the right click menu.
2. Check the Name says Credit Check.
3. Change the Display Name for the successful node from Default to Passed. Click OK to return to the Workspace. Please check the node names on the tooltips for this Action Agent to make sure that you have change the name of the node correctly.
4. Now you are going to add the details for the Inform node. To do this double click on the next line down in the grid box.



Warning: Do not be alarmed when you see three rows, all with 1st Rule in the Combination column. This is correct. Please continue.

5. There are two conditions to enter for this node. The customer is to be informed if the CUSTOMER CREDIT USAGE is greater than, or equal to, 75 and less than, or equal to, 100. Complete the grid lines as shown below.

Properties for Action Agent

Name: Credit Check Document type: SALESORDER

Description: Server executed Action Agent to check the percentage of the available credit limit that a customer has used.

Title overrides: Credit Checking is now taking place.

Define exit node rules

Code	Combination	Input Condition	Operator	Test Condition	Display Name
01	1st Rule	Customer Credit Usage Percentage	>	100	Fail
(Default)					Passed
02	1st Rule	Customer Credit Usage Percentage	>=	75	Inform
02	And	Customer Credit Usage Percentage	<=	100	Inform

☐ Save definition to palette on exit

OK Cancel

6. Please note that each node is defined by the Code number, and in this case the Inform node is Code number 02. Every rule has a new row and there may be more than one rule for each node, as there is in this example where Node 02 has two rules. Constructor completes the Combination column for you so that the first row for each Code is defined as 1st Rule and any subsequent rules are defined as And.



Tip: When you check these nodes on the Workspace you will find that the new Inform node is correctly positioned above the Fail node, very conveniently for you to add the new processing above and between Credit Check and Picking Notes. You might have expected it to be placed at the bottom to match the order in the list, however, Constructor places nodes on symbols in an anti-clockwise order starting at the right.

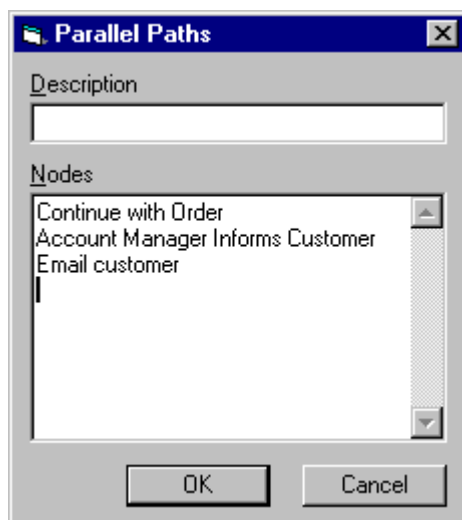
Step 3 Add A Parallel Path

In this example, when a customer is near their credit limit there are three simultaneous actions which you would like to carry out:

- You can Email the customer
- You can ask the Account Manager to inform the customer
- You can continue with the order.

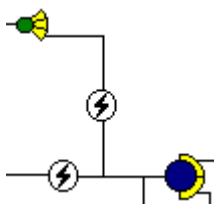
You are now going to reflect this by branching the processing from the Inform node into three pathways which are going to be processed simultaneously. This is what to do:

1. Drag a Parallel Path symbol from the Controls folder onto the Workspace.
2. Select Edit from the right click menu and amend the node details as shown below. Please delete the description. These also become the tooltips for each node:



3. Check the nodes on the Workspace by holding the cursor over each one. If they are in the wrong order then please re-order them using the Resequence Links option on the right click menu.
4. Join Credit Check Action Agent (Inform Node) to the Parallel Path symbol.
5. Join the bottom node (Continue With Orders) on the Parallel Path to the Picking Note symbol. Constructor will join the line slightly to the left of the Picking Note symbol.
6. Check that you have an Automatic Immediate scheduling symbol before Picking Note on the line from the Credit Check Action Agent.
7. Insert an Automatic Immediate scheduling symbol before Picking Note on the line from the Parallel Path.

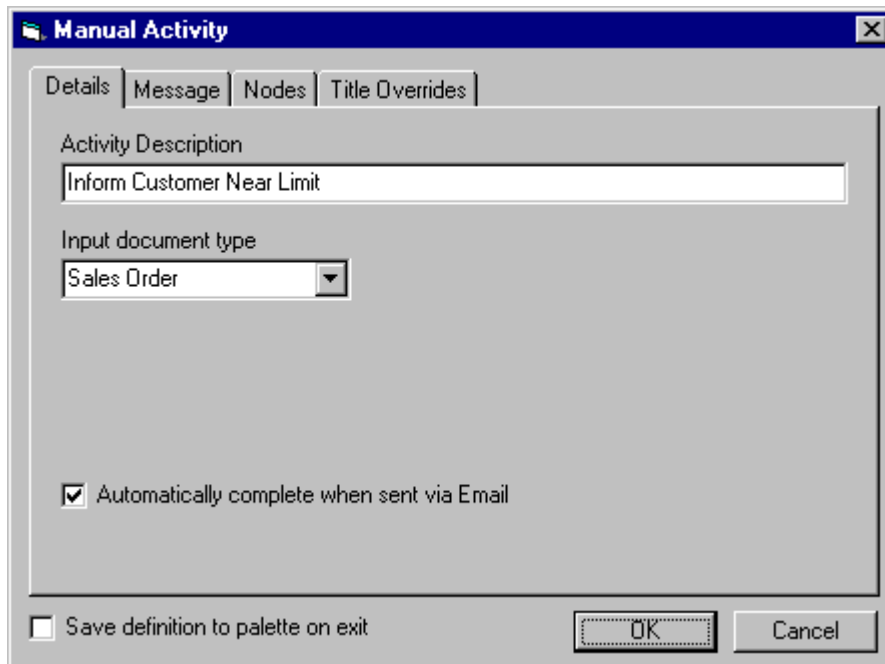
This is what you get:



Step 4 Add Two Manual Activities

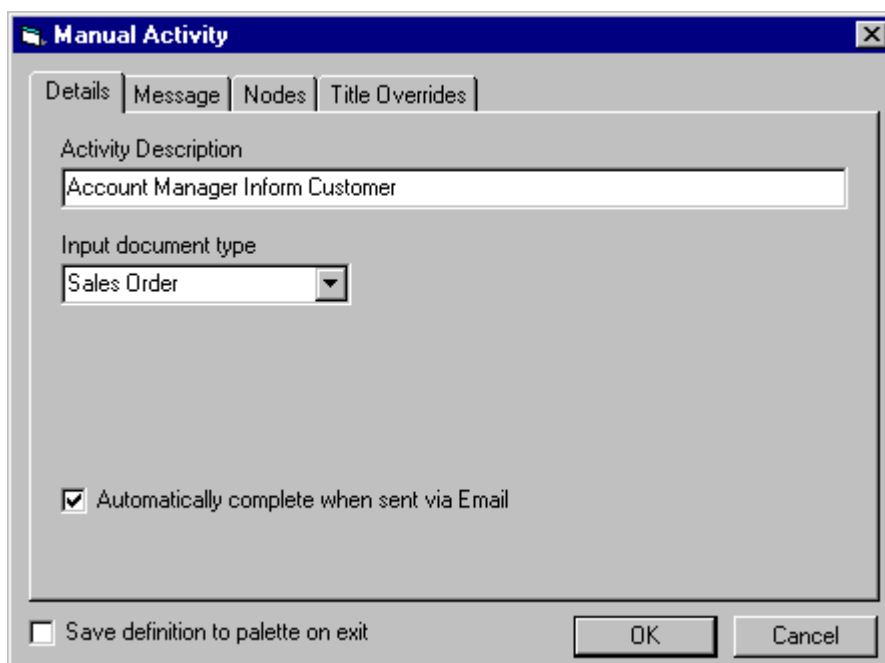
The top two nodes from the Parallel Path are going to be handled by Manual Activities. This is what to do:

1. Drag a Manual Activity from the Controls folder on to the Workspace.
2. The Manual Activity Window opens immediately. Change the Activity Description to Inform Customer Near Limit. Select Sales Order for the Input Document Type and click OK.



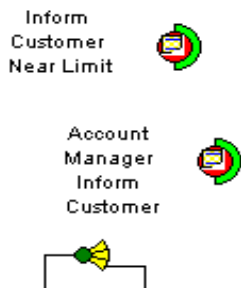
The image shows a 'Manual Activity' dialog box with a blue title bar and a close button. It has four tabs: 'Details', 'Message', 'Nodes', and 'Title Overrides'. The 'Details' tab is selected. Inside the dialog, there is a text field for 'Activity Description' containing 'Inform Customer Near Limit'. Below it is a dropdown menu for 'Input document type' with 'Sales Order' selected. At the bottom left, there is a checked checkbox labeled 'Automatically complete when sent via Email'. At the bottom right, there are 'OK' and 'Cancel' buttons. At the bottom left, there is an unchecked checkbox labeled 'Save definition to palette on exit'.

3. Drag a second Manual Activity from the Controls folder onto the Workspace.
4. The Manual Activity Window opens immediately. Change the Activity Description to Account Manager Inform Customer. Select Sales Order for the Input Document Type and click OK.

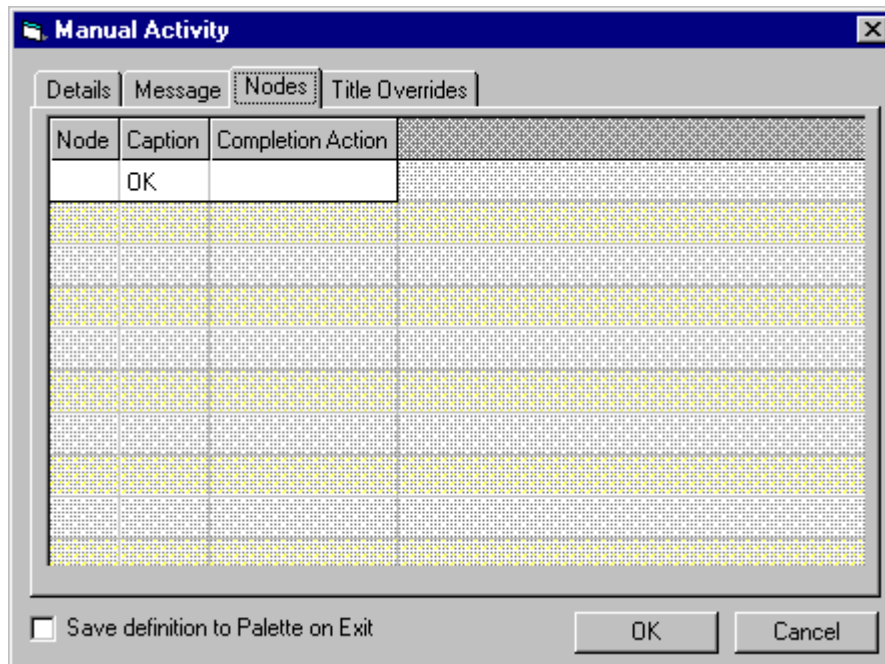


The image shows a 'Manual Activity' dialog box with a blue title bar and a close button. It has four tabs: 'Details', 'Message', 'Nodes', and 'Title Overrides'. The 'Details' tab is selected. Inside the dialog, there is a text field for 'Activity Description' containing 'Account Manager Inform Customer'. Below it is a dropdown menu for 'Input document type' with 'Sales Order' selected. At the bottom left, there is a checked checkbox labeled 'Automatically complete when sent via Email'. At the bottom right, there are 'OK' and 'Cancel' buttons. At the bottom left, there is an unchecked checkbox labeled 'Save definition to palette on exit'.

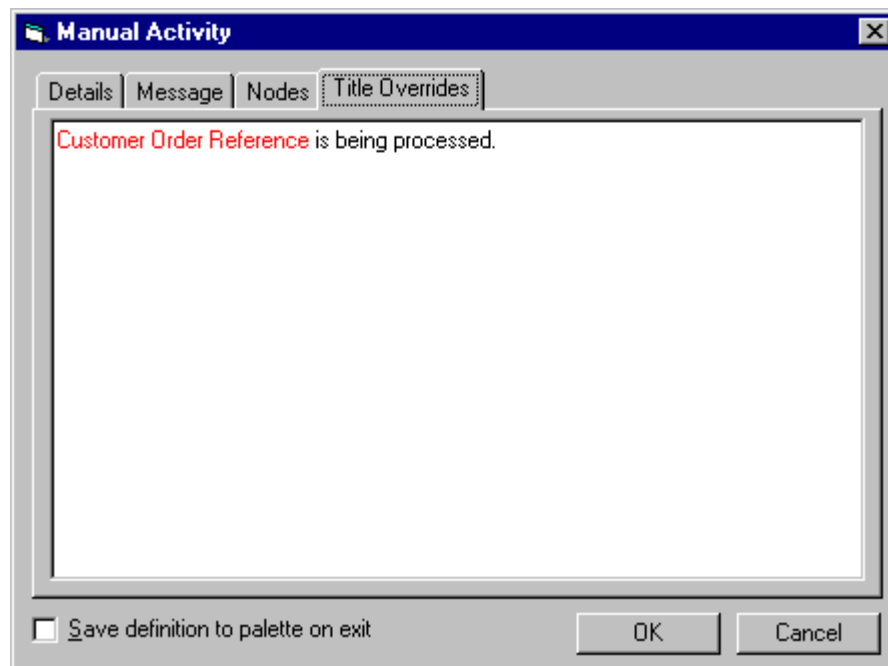
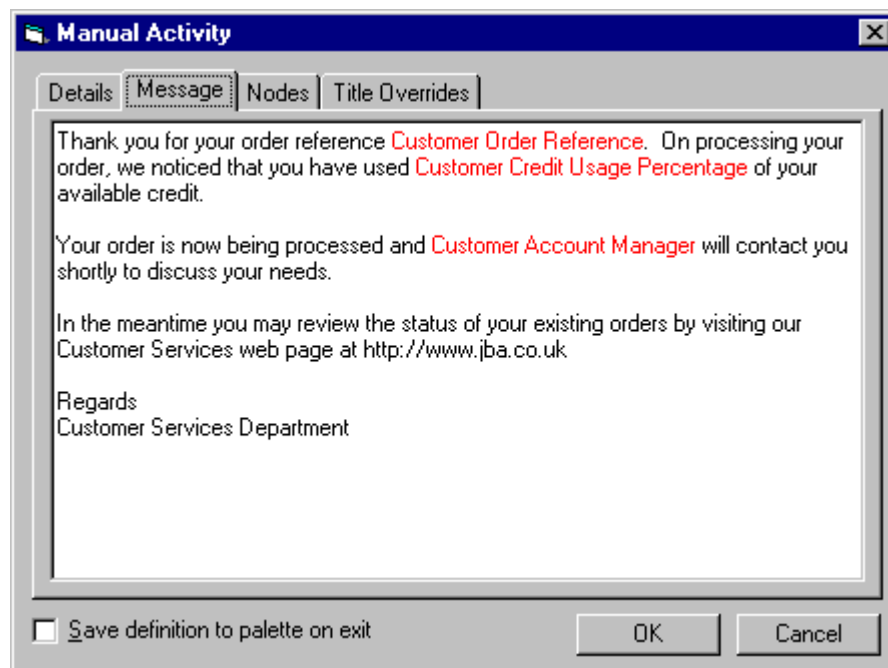
This is how it looks on the Workspace:



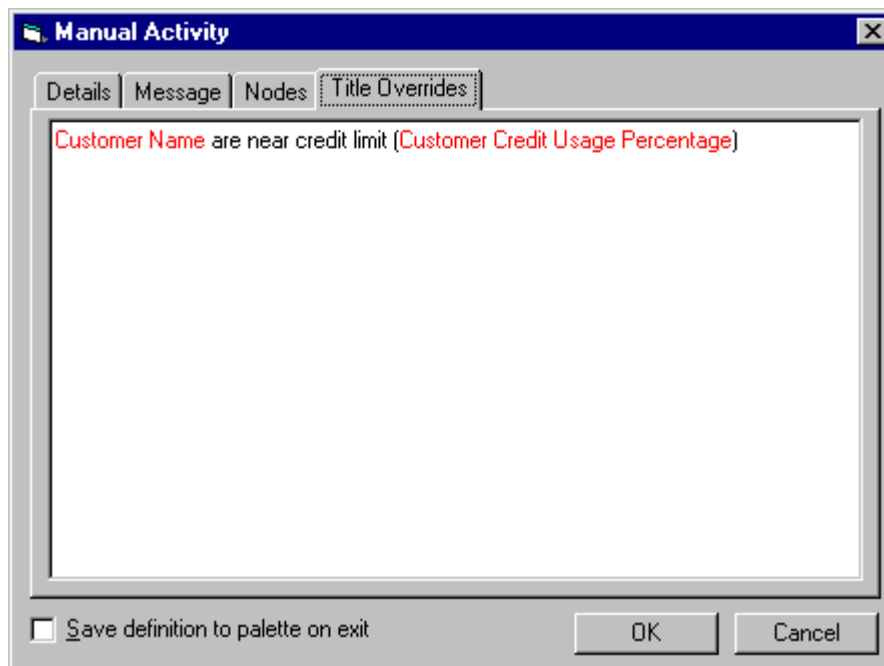
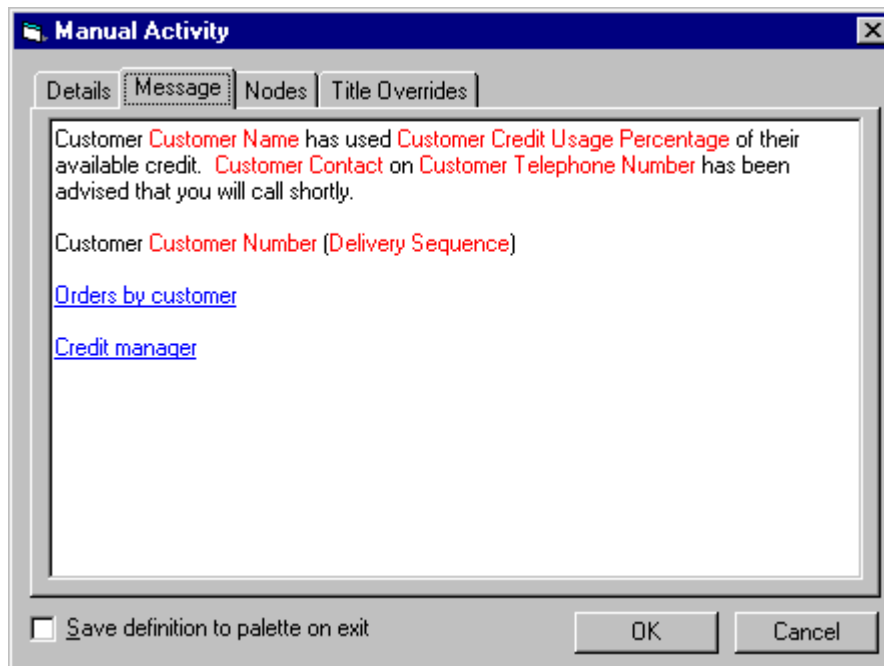
5. Note that the Manual Activity Window does not show any node details because you haven't connected the symbols yet.
6. Join up the top node from the Parallel Path, called Email Customer, to the Manual Activity called Inform Customer Near Limit.
7. Join up the middle node from the Parallel Path, called Account Manager Informs Customer, to the Manual Activity called Account Manager Inform Customer.
8. Note that in both cases the Nodes panel has one node for OK.



9. Now complete the Message and Title Overrides panels for Inform Customer Near Limit, as follows:

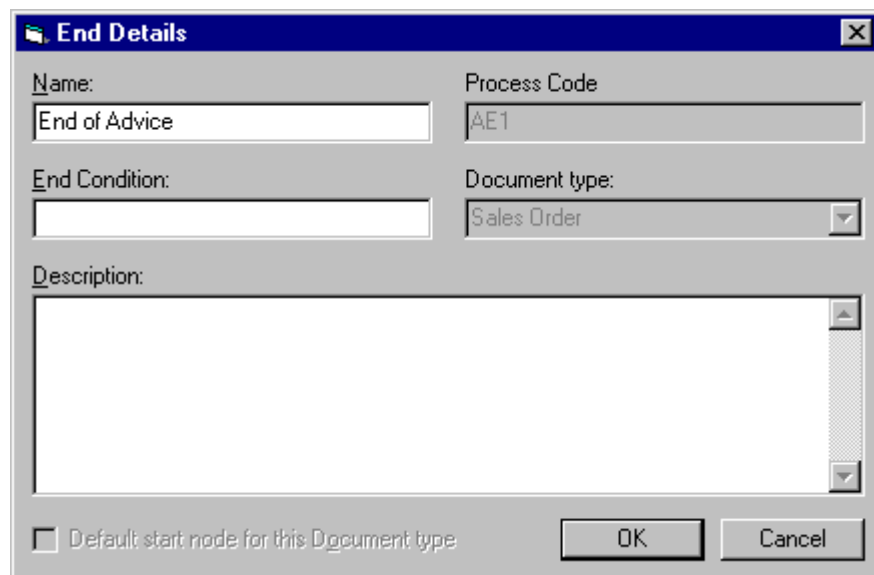


10 Now complete the Message and Title Overrides panels for Account Manager Inform Customer, as follows:



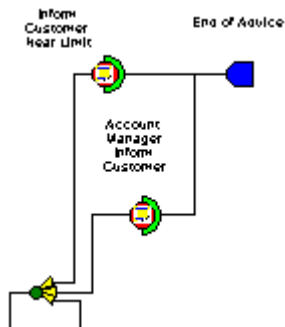
Step 5 Add The End Condition

1. Drag an End Condition out on to the Workspace.
2. Select Edit from the right click menu and change the name of the End Condition to End of Advice.

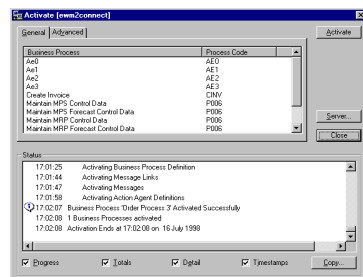


The 'End Details' dialog box is shown. It has a title bar with a close button. The 'Name' field contains 'End of Advice'. The 'Process Code' field contains 'AE1'. The 'End Condition' field is empty. The 'Document type' dropdown menu is set to 'Sales Order'. There is a large 'Description' text area below. At the bottom, there is a checkbox labeled 'Default start node for this Document type' which is unchecked, and 'OK' and 'Cancel' buttons.

3. Select Colour from the right click menu and change the colour of the end symbol to blue.
4. Join the two Manual Activities to this End Condition. This is how it should look on the Workspace:



5. Note that Steps 6 and 7 explain how to go on and add schedules for these Manual Activities. This is what you will be aiming for:



Step 6 Add User Schedule

Warning: Please make sure that you have followed the Set Up Users instructions in the Before You Start section of this help guide.

Now you are going to add scheduling details for the Account Manager Inform Customer activity.

This involves passing an activity on to another user so you set up a User Schedule. The activity will then be passed on to the Account Manger and will appear in the Action List for any System 21 Explorer user with the Role of Account Manager.

Check Data Field

1. First check that you have a data field defined as Customer Account Manager. Select Data Fields from the Work Management Menu. This displays the Data Field Window.

Document type	Description	Field type	Usage	Data field information
PICKNOTE	Customer Account Manager	S	UD	Select PRMD15 from osld1f3/INP...
SALESORDER	Admin E-mail	T	E	z1admin@z1.co.uk
	Acknowledge Print Flag	S	D	Select PRNA40 from OEP40 wher...
	Activity	W	D	
	Agent	S	D	Select AGNT40 from OEP40 wher...
	Base Order Value	S	D	Select BORV40 from OEP40 wher...
	BusinessObject	W	D	
	BusinessProcess	W	D	
	Carrier	S	D	Select CARR40 from OEP40 wher...
	Company	W	D	
	Currency Code	S	D	Select CURN40 from OEP40 wher...
	Currency Rate	S	D	Select CURT40 from OEP40 wher...
	Currency Rate Code	S	D	Select CURC40 from OEP40 wher...
	CurrentDate	W	D	
	CurrentTime	W	D	
	Customer Accompanying Invoice Flag	S	D	Select AINF20 from OEP40,OEP2...
	Customer Acknowledgment	S	D	Select ACKN20 from OEP40,OEP2...
	Customer Address Line 1	S	D	Select CAD105 from OEP40,SLP0...
	Customer Address Line 2	S	D	Select CAD205 from OEP40,SLP0...
	Customer Address Line 3	S	D	Select CAD305 from OEP40,SLP0...
	Customer Address Line 4	S	D	Select CAD405 from OEP40,SLP0...
	Customer Address Line 5	S	D	Select CAD505 from OEP40,SLP0...
	Customer Agent	S	D	Select AGNT20 from OEP40,OEP...

Buttons: Add... Edit... Delete Close

2. Select a Document Type of Sales Order and look for Customer Account Manager.
3. Note that this field has a Usage of U for User because it is a User schedule and D because it is also a Data Field.
4. If the Customer Account Manager does not exist select the Add button and add the fields. The Window below shows how to enter new details for Customer Account Manager.

Edit Data Field

Data field name: CustAcMgr Document type: SALESORDER Data field type: SQL

Description: Customer Account Manager

SQL statement:
Select PRMD15 from osld1f3/INP15, osld1f3/OEP40 where CONO40 = CONO15 and PRMT15 = 'SLMN' and PSAR15 = CUSN40 and CONO40 = Company and ORD40 = Document

Usage:
☐ Role
☒ User
☐ E-mail address
☐ Priority
☒ Description

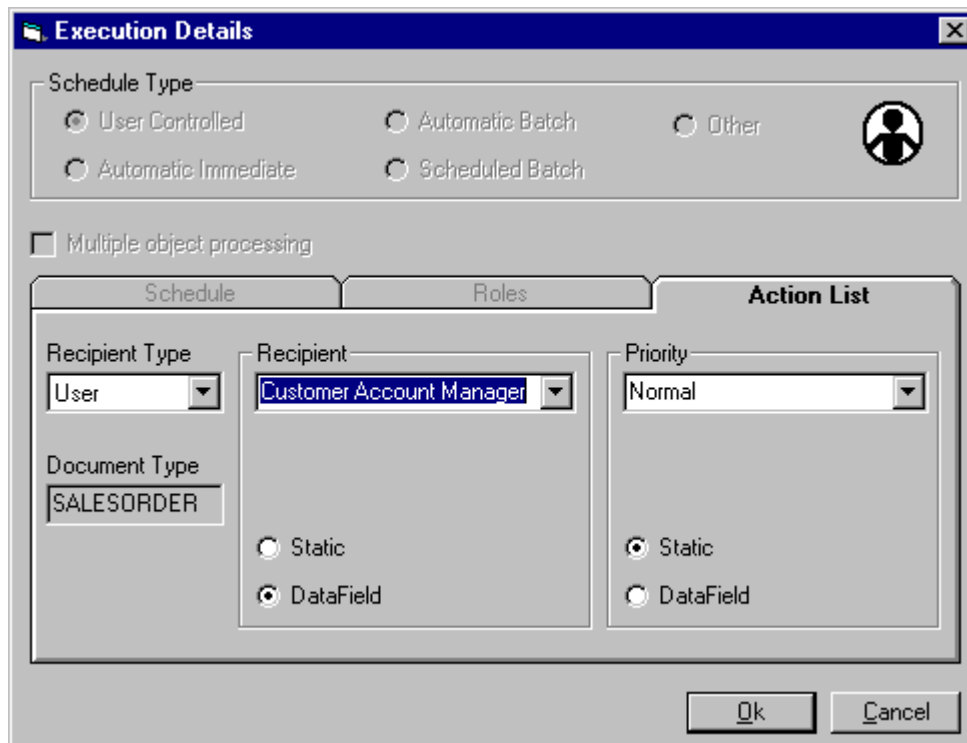
Data type: Alpha string Field length: 30 Decimal places: 0 OK Cancel



Warning: Do not edit a Data Field with a Data Field Type of Workflow or Programmatic, because this will affect the design time database. You can edit Text or SQL Data Fields.

Enter Schedule

1. Place the cursor over the connection line going into the Account Manager Inform Customer activity.
2. Select Edit from the right click menu. Constructor recognises that you have a Manual Activity and displays the Execution Details Window with the Action List panel ready for you.
3. Note that Recipient uses Data Field and not Static option. Use static only when you want to type in a user name, for example, John Smith.
4. Select User for the Recipient Type and Customer Account Manager for Recipient.
5. Check your details with the window below and select OK. Constructor puts a User symbol on the line for you.



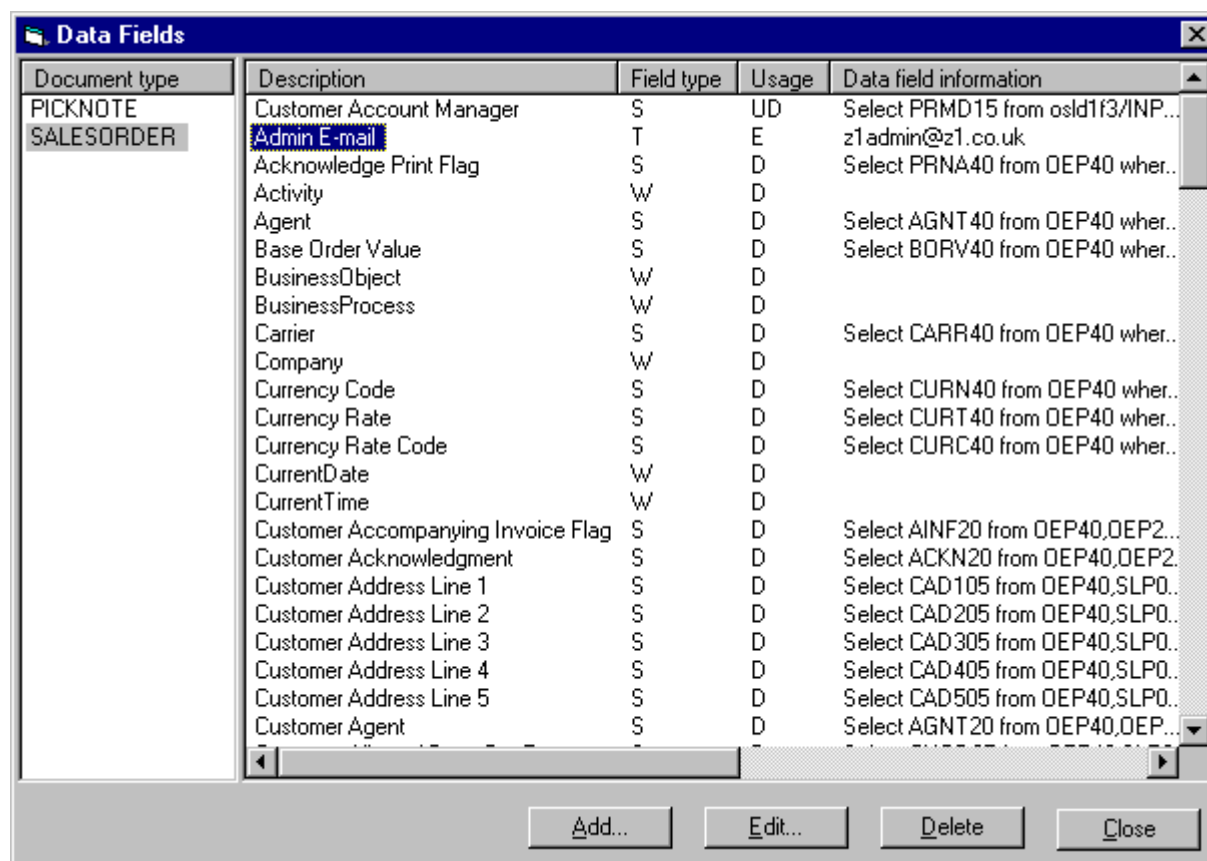
The **Execution Details** dialog box is used to configure the execution of an action. It features a title bar with a close button. The main area is divided into several sections:

- Schedule Type:** A group box containing four radio buttons: **User Controlled** (selected), **Automatic Batch**, **Other**, and **Automatic Immediate**. A user icon is located to the right of these options.
- Multiple object processing:** A checkbox that is currently unchecked.
- Tabbed Interface:** Three tabs are visible: **Schedule**, **Roles**, and **Action List**. The **Schedule** tab is active.
- Schedule Tab:**
 - Recipient Type:** A dropdown menu showing **User**.
 - Document Type:** A text box containing **SALESORDER**.
 - Recipient:** A dropdown menu showing **Customer Account Manager**.
 - Static/DataField:** Two radio buttons, with **DataField** selected.
- Action List Tab:**
 - Priority:** A dropdown menu showing **Normal**.
 - Static/DataField:** Two radio buttons, with **Static** selected.
- Buttons:** **Ok** and **Cancel** buttons are located at the bottom right.

Step 7 Add Email Schedule

Now you are going to add scheduling details for the Inform Customer Near Limit activity.

This will send a message by Email to the customer to advise them that they are near their credit limit.



Check Data Field

1. First you need a data field for the Email address. Select Data Fields from the Work Management Menu. This displays the Data Field Window.
2. Select Sales Order in the Document Type column and check for a Data Field with a Description of Admin Email. If the Data Field does not exist then you may add it. You would select the Add button to display the New Data Field Window and enter the details as shown below:

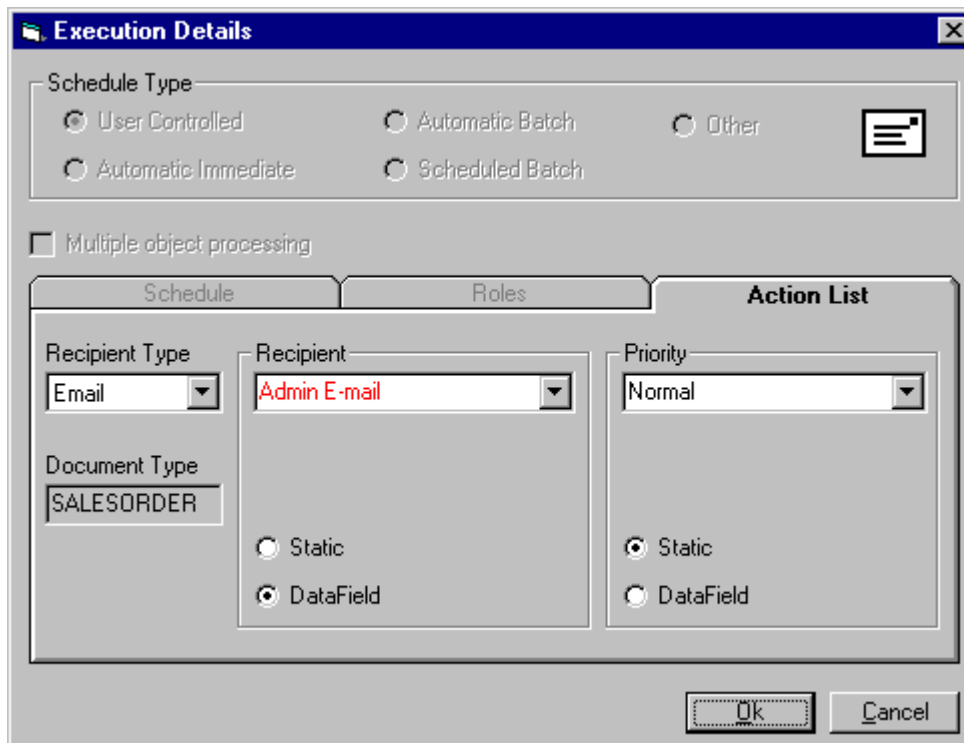
3. Note that this field has a Usage of E for Email.
4. Note that it is the Description which you use next for the Recipient in the Execution Details window and not the Name.



Warning: Do not edit a Data Field with a Data Field Type of Workflow or Programmatic because this will affect the design time database. You can edit Text or SQL Data Fields.

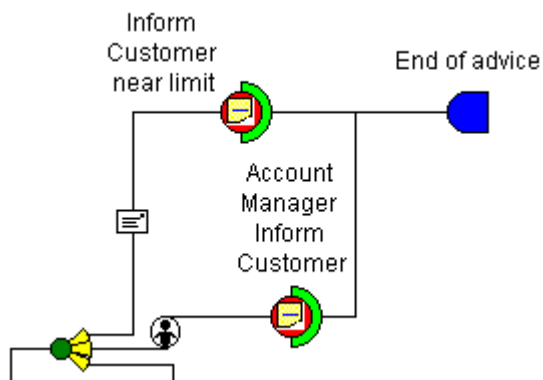
Enter Schedule

1. Place the cursor over the connection line going into the Inform Customer Near Limit activity.
2. Select Edit from the right click menu. Constructor recognises that you have a Manual Activity and displays the Execution Details Window with only the Action List panel ready for you.
3. Select Email for the Recipient Type and Admin Email (this was the Description on the Data Fields Window) for Recipient.
4. Note that Recipient uses the Data Field and not the Static option. Use static only when you want to type in a Web address manually just for this particular Business Process. As the data field has been set up you can use it as often as you like.
5. Check your details with the window below and select OK. Constructor puts an Email symbol on the line for you.



The **Execution Details** dialog box is used to configure the execution of a workflow. It features a **Schedule Type** section with radio buttons for **User Controlled** (selected), **Automatic Batch**, **Other**, **Automatic Immediate**, and **Scheduled Batch**. A **Multiple object processing** checkbox is also present. Below these are three tabs: **Schedule**, **Roles**, and **Action List**. The **Schedule** tab is active, showing a **Recipient Type** dropdown set to **Email**, a **Document Type** dropdown set to **SALESORDER**, a **Recipient** dropdown set to **Admin E-mail**, and a **Priority** dropdown set to **Normal**. Under the **Recipient** dropdown, there are radio buttons for **Static** and **DataField** (selected). Under the **Priority** dropdown, there are radio buttons for **Static** (selected) and **DataField**. At the bottom are **Ok** and **Cancel** buttons.

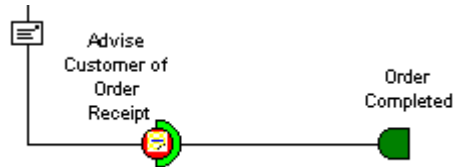
6. This is how the scheduling details look on the Workspace:



Step 8 One Last Email

There is one last thing to do and that is to add an Email symbol before Advise Customer Of Order Receipt.

This is what you are aiming for:



Warning: Please also go back to your Order Process 1 and Order Process 2 and put the same Email symbol on the connection line leading into the Advise Customer Of Order Receipt symbol. In Ae1 and Ae2 we recommended that you leave this until Ae3 and this is the time now to go back and do it.

Please enter these details as follows:

The Business Process Ae3 is now complete.

Remember to save Order Process 1, Order Process 2 and Order Process 3.

You can close everything except Order Process 3.

Now refer to the help text for Using Validate And Walkthrough Buttons.

Pre-activation Checking



Warning: *Don't be tempted to miss out this stage.*

Introduction

You will shortly be activating your Business Process. This means uploading it to the server so that Work Management can then run it using the Work Management Engine. The activation process involves rigorous validation checking and if it fails you will have to make corrections and then start the activation again. It is much easier, and safer, to check it thoroughly first.

There are two buttons on the toolbar that you can use, called Validate and Walkthrough. Validate performs the same validation checks as the Activation program.

Please check these now:

1. Check that all symbols on the Workspace are connected.
2. Check that all activities, except Initial Activities, have Execution Modes leading into them.
- 3 Check that Document Types are correctly defined. This is explained in more detail in the next section.

When you are sure that your design is error free you can use the Walkthrough button to follow the logical flow of the program as the user would see it.



Note: You can validate and Walkthrough a design at any stage. With experience, you may prefer to do this at logical stages of the building process to test it. However, for the purposes of this tutorial we explain this once now after building Ae3.

About Document Types

Document Types have to be correct and follow a logical pathway, and nodes have to be correctly linked to activities.



Tip: Please read the notes below and then follow the examples until you feel confident about how to check Document Types and Nodes. If you have made any mistakes then tasks outlined in the Validation section, next in the tutorial, will highlight them for you.

Activities

Go through your Business Process and for each Activity check the Document Type going in and the node(s) going out. To do this you select the Properties option from the right click menu. This displays the Properties Window.

Look at the General panel to find the Input Document Type. There is a list of these so you have to make sure that the Document Type you have selected matches the Document Type entered against the output node from the previous Activity.

Look at the Details panel to find the Output Nodes. For each node check the Description and the Document Type and then check that they are joined up correctly to the next activities. Hold the cursor over each node to display its name: this matches the Description on the Details panel.

Manual Activities

Go through your Business Process and for each Manual Activity check the Document Type going in and the nodes going out. To do this you select the Properties option from the right click menu. This displays the Manual Activity Window.

Look at the Details panel to find the Input Document Type. There is a list of these so you have to make sure that the Document Type you have selected matches the Document Type entered against the output node from the previous Activity.

Look at the Nodes panel to find the Output Nodes. For each node check the Caption and then check that they are joined up correctly to the next activities. Hold the cursor over each node to display its name: this matches the Caption on the Manual Activity Window. Unlike an Activity, a Manual Activity takes the Input Document Type, which you selected, and passes it straight through so here you do not have to enter the Document Type against the output nodes.

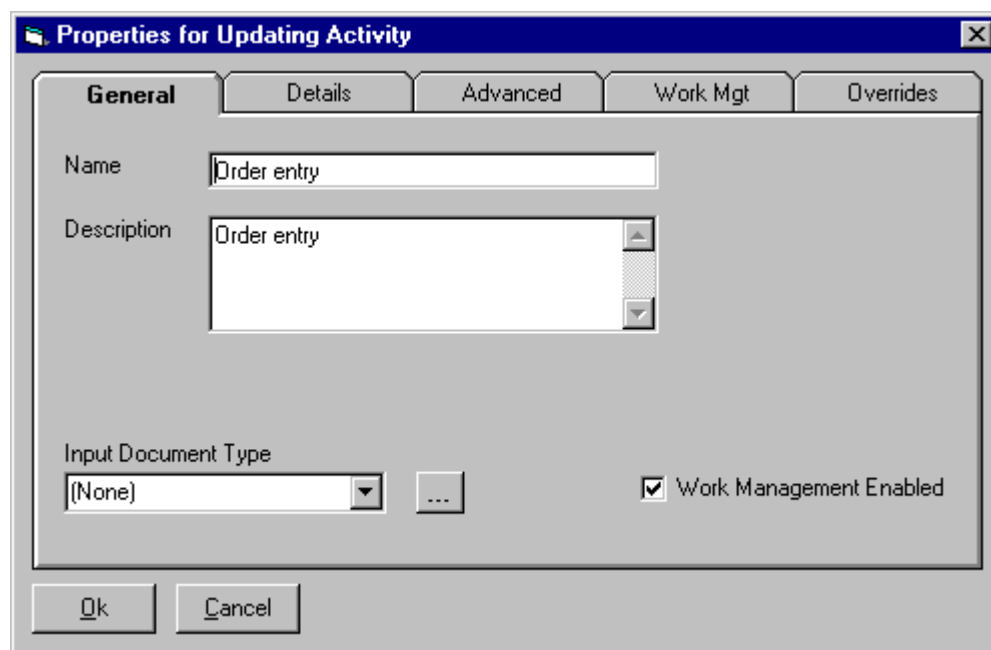
Action Agents

When you select the Properties option from the right click menu this displays the Action Agent Path Window.

Unlike an Activity, an Action Agent takes the Document Type from the previous Activity and passes it straight through, so you cannot enter an incorrect Document Type here. However, you should check the name of the Document Type to make sure the correct one is being passed through. Then check on the Workspace that the nodes are joined up correctly. Hold the cursor over each node to display its name: this matches the Display Name on the Action Agent Path Window.

Examples

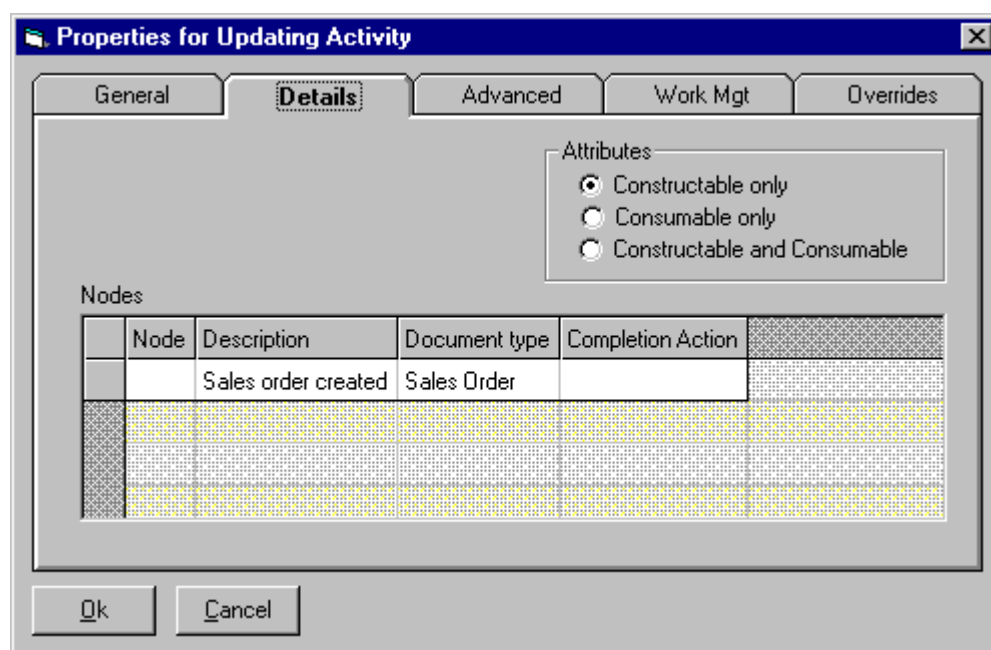
Look at Order Entry and note, on the General panel, that there is no Input Document Type because this is the start of the process:



The 'Properties for Updating Activity' dialog box, General tab. It contains fields for Name and Description, both set to 'Order entry'. The Input Document Type is set to '(None)'. There is a checkbox for 'Work Management Enabled' which is checked. Buttons for 'Ok' and 'Cancel' are at the bottom.

Tab	Name	Description	Input Document Type	Work Management Enabled
General	Order entry	Order entry	(None)	☒

However, by looking at the Details panel, you can see that there is one output node with a Sales Order Document Type:



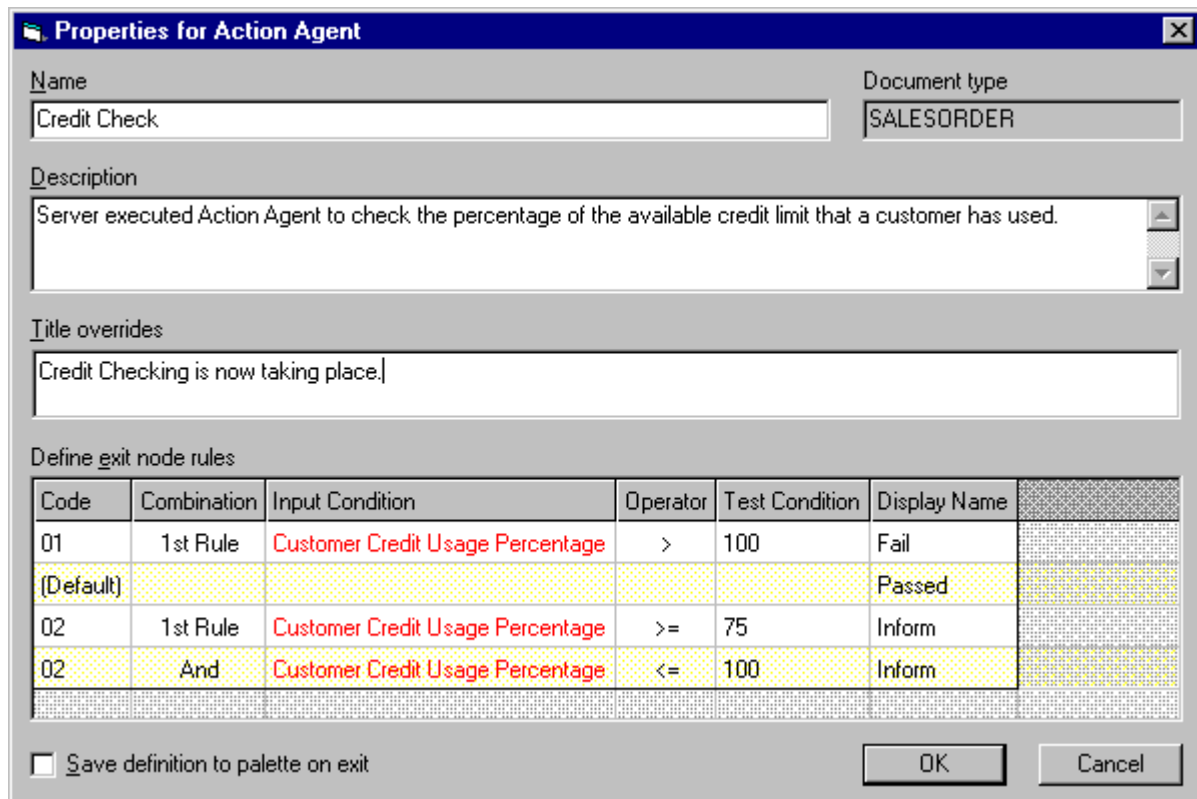
The 'Properties for Updating Activity' dialog box, Details tab. It shows a table of nodes. The first node is 'Sales order created' with a 'Sales Order' document type. There are also radio buttons for 'Attributes'.

Node	Description	Document type	Completion Action
	Sales order created	Sales Order	

Attributes:

- ☒ Constructable only
- ☐ Consumable only
- ☐ Constructable and Consumable

Look at Credit Check Action Agent and note that the Document Type is Sales Order. The field name does not say Input Document Type: it says only Document Type. This is because an Action Agent always takes the Document Type from the previous activity and passes it out again.



Properties for Action Agent

Name: Credit Check

Document type: SALESORDER

Description: Server executed Action Agent to check the percentage of the available credit limit that a customer has used.

Title overrides: Credit Checking is now taking place.

Define exit node rules

Code	Combination	Input Condition	Operator	Test Condition	Display Name
01	1st Rule	Customer Credit Usage Percentage	>	100	Fail
(Default)					Passed
02	1st Rule	Customer Credit Usage Percentage	>=	75	Inform
02	And	Customer Credit Usage Percentage	<=	100	Inform

☐ Save definition to palette on exit

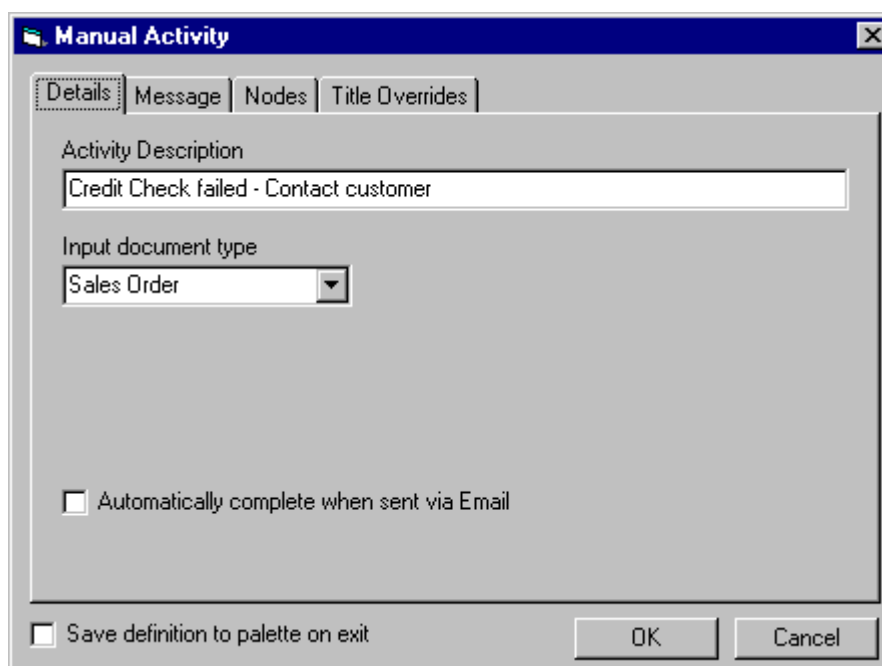
OK Cancel

This means that from the Credit Check Sales Order becomes the Document Type going forward along all three pathways: Inform, Passed and Fail.

Inform and Fail nodes

On the pathways from the Inform node and the Fail node, Sales Order is passed all the way along the lines to the End Conditions.

For example: look at the Manual Activity called Credit Check Failed - Contact Customer. The Input Document Type is Sales Order



Manual Activity

Details Message Nodes Title Overrides

Activity Description: Credit Check failed - Contact customer

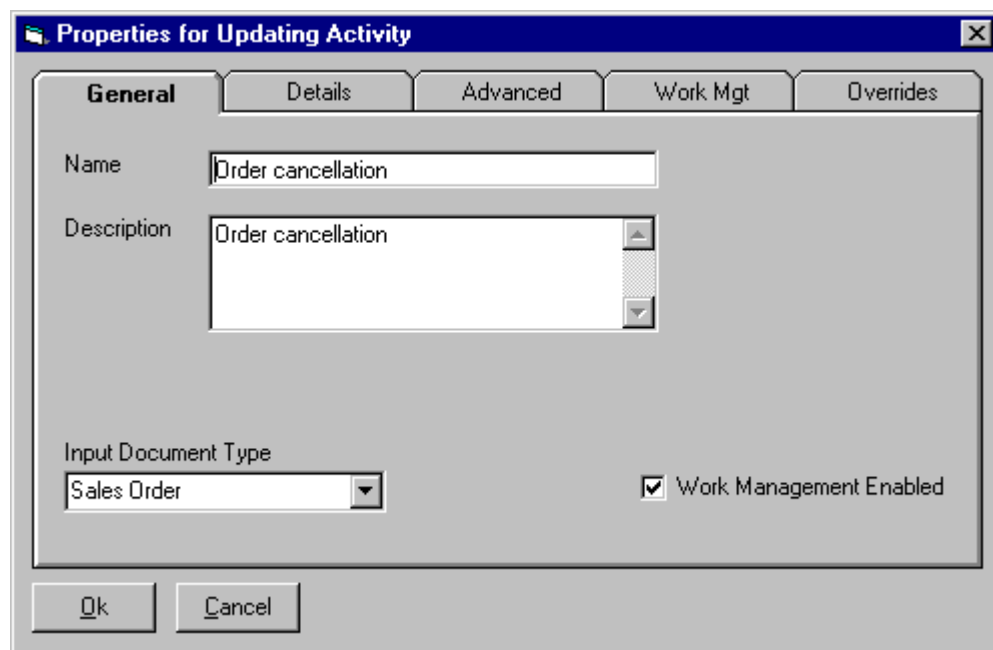
Input document type: Sales Order

☐ Automatically complete when sent via Email

☐ Save definition to palette on exit

OK Cancel

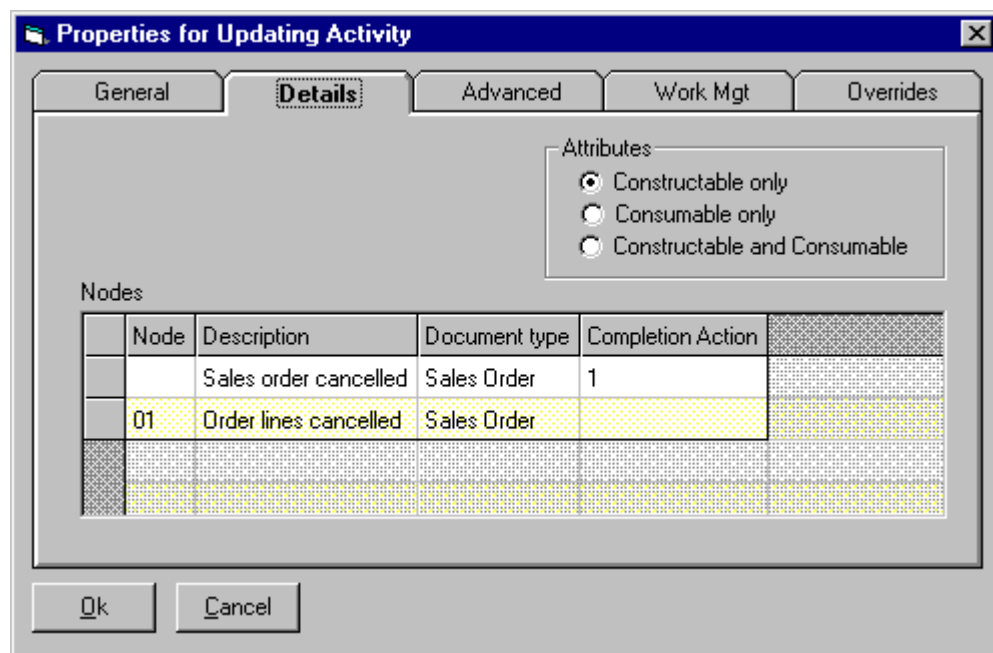
Also look at the Activity called Order Cancellation. The Input Document Type is Sales Order, as shown on the General panel.



The screenshot shows the 'Properties for Updating Activity' dialog box with the 'General' tab selected. The 'Name' field contains 'Order cancellation'. The 'Description' field contains 'Order cancellation'. The 'Input Document Type' dropdown menu is set to 'Sales Order'. The 'Work Management Enabled' checkbox is checked. The 'Ok' and 'Cancel' buttons are at the bottom.

Tab	Name	Description	Input Document Type	Work Management Enabled
General	Order cancellation	Order cancellation	Sales Order	☒

And the Output Nodes also pass a Sales Order Document Type on to the next activity.

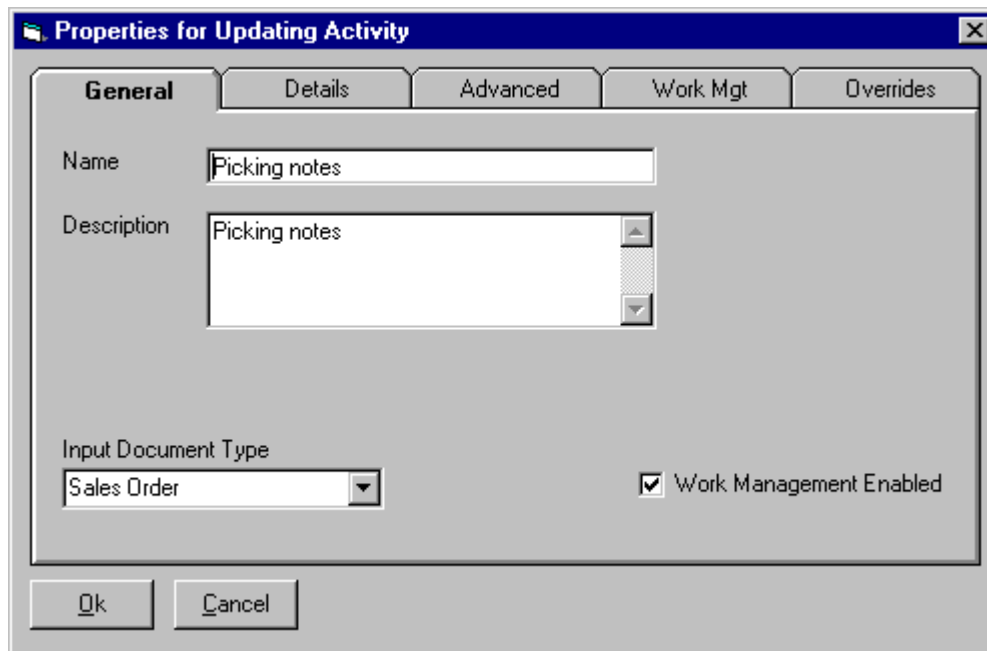


The screenshot shows the 'Properties for Updating Activity' dialog box with the 'Details' tab selected. The 'Attributes' section has three radio buttons: 'Constructable only' (selected), 'Consumable only', and 'Constructable and Consumable'. The 'Nodes' section contains a table with two rows. The first row has a blank 'Node' field, 'Sales order cancelled' description, 'Sales Order' document type, and '1' completion action. The second row has '01' as the node, 'Order lines cancelled' description, 'Sales Order' document type, and a blank completion action.

Tab	Attributes	Nodes												
Details	☒ Constructable only ☐ Consumable only ☐ Constructable and Consumable	<table border="1"><thead><tr><th>Node</th><th>Description</th><th>Document type</th><th>Completion Action</th></tr></thead><tbody><tr><td></td><td>Sales order cancelled</td><td>Sales Order</td><td>1</td></tr><tr><td>01</td><td>Order lines cancelled</td><td>Sales Order</td><td></td></tr></tbody></table>	Node	Description	Document type	Completion Action		Sales order cancelled	Sales Order	1	01	Order lines cancelled	Sales Order	
Node	Description	Document type	Completion Action											
	Sales order cancelled	Sales Order	1											
01	Order lines cancelled	Sales Order												

Passed Node

The Document Type changes when it reaches the Picking Notes activity, where the Input Document Type is Sales Order.



Properties for Updating Activity

General | Details | Advanced | Work Mgt | Overrides

Name: Picking notes

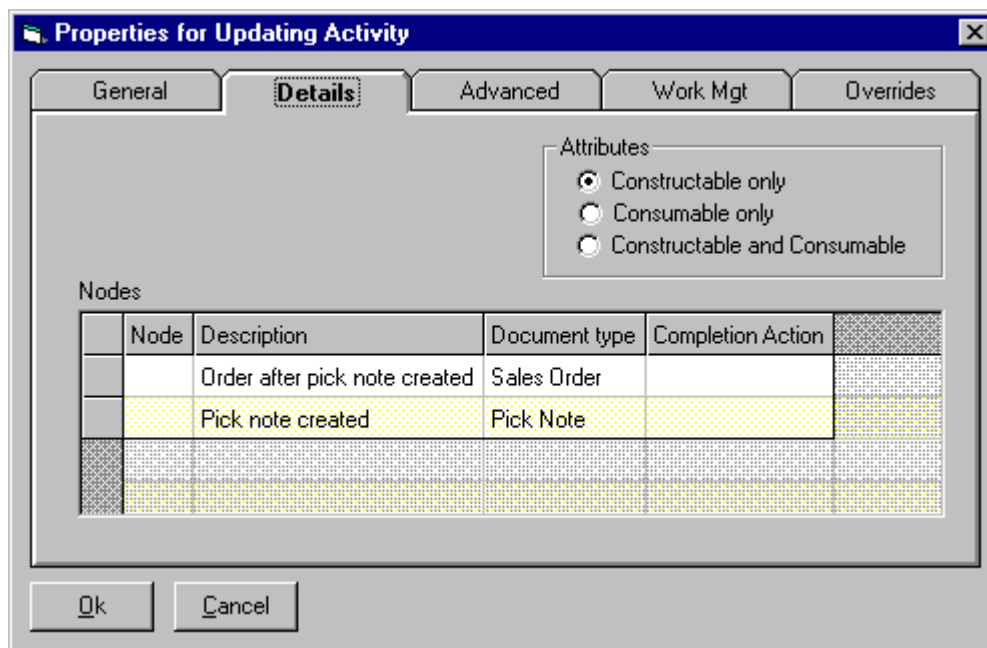
Description: Picking notes

Input Document Type: Sales Order

☒ Work Management Enabled

Ok Cancel

However, one of the output nodes has a Pick Note Document Type which then goes on to Confirmation Of Despatch.



Properties for Updating Activity

General | **Details** | Advanced | Work Mgt | Overrides

Attributes

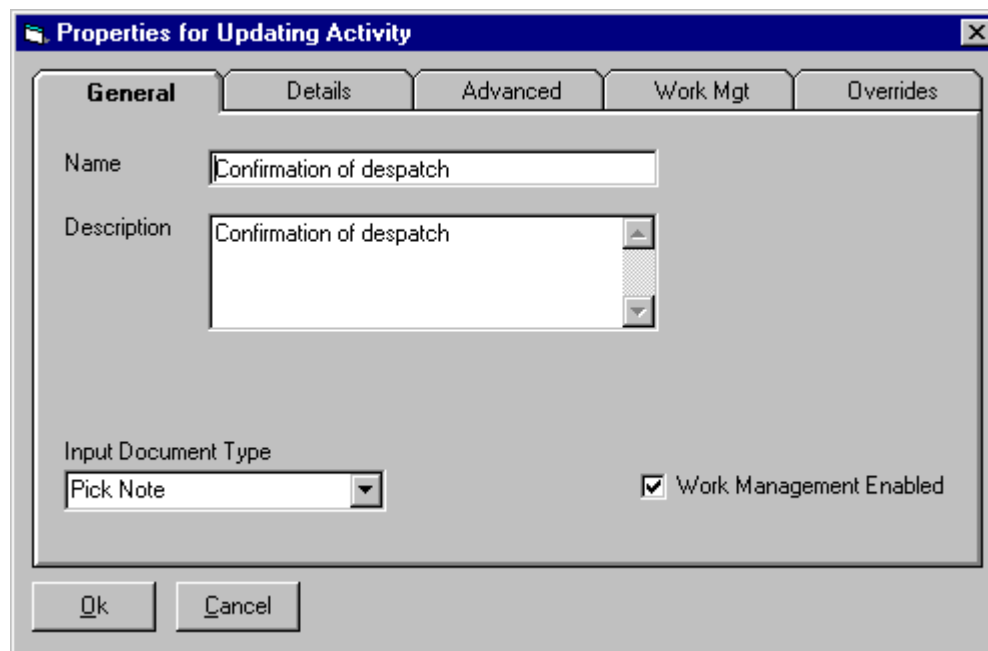
- ☒ Constructable only
- ☐ Consumable only
- ☐ Constructable and Consumable

Nodes

Node	Description	Document type	Completion Action
	Order after pick note created	Sales Order	
	Pick note created	Pick Note	

Ok Cancel

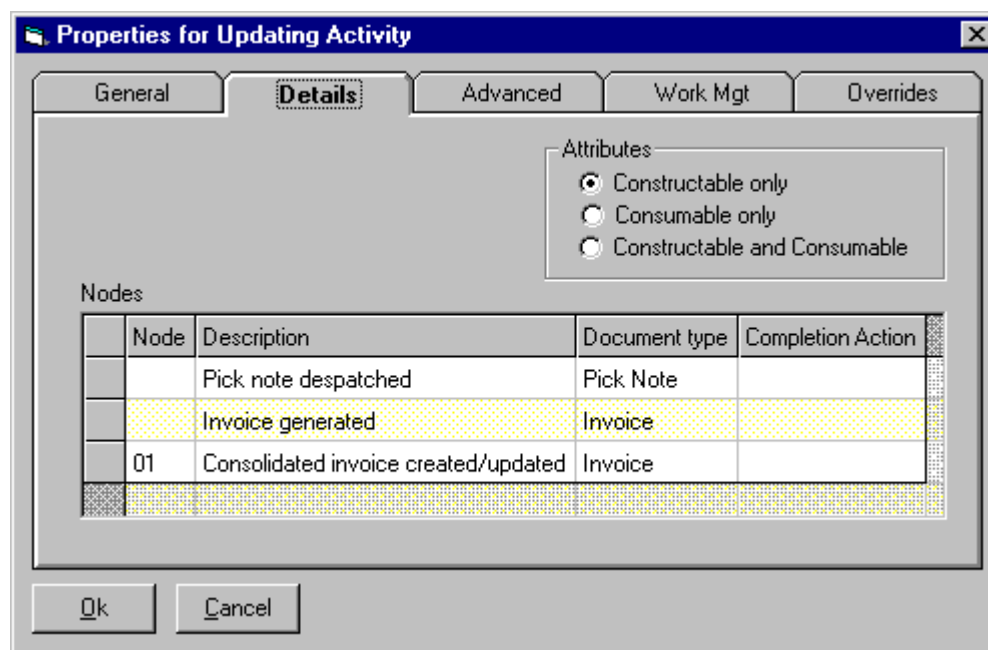
Note then that Confirmation Of Despatch has an Input Document Type of Pick Note:



The 'Properties for Updating Activity' dialog box, General tab. It contains fields for Name and Description, both set to 'Confirmation of despatch'. The Input Document Type is set to 'Pick Note'. A checkbox for 'Work Management Enabled' is checked. Buttons for 'Ok' and 'Cancel' are at the bottom.

Tab	Name	Description	Input Document Type	Work Management Enabled
General	Confirmation of despatch	Confirmation of despatch	Pick Note	☒

However only one output node passes on a Pick Note Document Type and the other two output nodes both pass on an Invoice Document Type.



The 'Properties for Updating Activity' dialog box, Details tab. It shows attributes and a table of nodes. The 'Attributes' section has three radio buttons: 'Constructable only' (selected), 'Consumable only', and 'Constructable and Consumable'. The 'Nodes' table lists three nodes: 'Pick note despatched' (Pick Note), 'Invoice generated' (Invoice), and '01 Consolidated invoice created/updated' (Invoice).

Attributes				
☒	Constructable only			
☐	Consumable only			
☐	Constructable and Consumable			
Nodes				
Node	Description	Document type	Completion Action	
	Pick note despatched	Pick Note		
	Invoice generated	Invoice		
01	Consolidated invoice created/updated	Invoice		

The Print Invoices and Invoice Posting activities then deal only with Invoice Document Types.

The screenshot shows the 'Properties for Updating Activity' dialog box with the 'General' tab selected. The 'Name' field contains 'Print invoices'. The 'Description' field also contains 'Print invoices'. The 'Input Document Type' dropdown is set to 'Invoice'. The 'Work Management Enabled' checkbox is checked. The 'Ok' and 'Cancel' buttons are at the bottom.

Properties for Updating Activity

General Details Advanced Work Mgt Overrides

Name: Print invoices

Description: Print invoices

Input Document Type: Invoice

☒ Work Management Enabled

Ok Cancel

The screenshot shows the 'Properties for Updating Activity' dialog box with the 'Details' tab selected. The 'Attributes' section has three radio buttons: 'Constructable only' (selected), 'Consumable only', and 'Constructable and Consumable'. The 'Nodes' section contains a table with 5 columns: Node, Description, Document type, Completion Action, and an empty column. The table has 3 rows of data.

Properties for Updating Activity

General **Details** Advanced Work Mgt Overrides

Attributes

- ☒ Constructable only
- ☐ Consumable only
- ☐ Constructable and Consumable

Nodes

Node	Description	Document type	Completion Action	
	Invoice printed	Invoice		
01	Failed invoice print	Invoice		

Ok Cancel

Please go on to the next section to validate your design.

Validating

Please check that you have Order Process 3 selected on the Workspace.

Select the Validate button on the Toolbar.



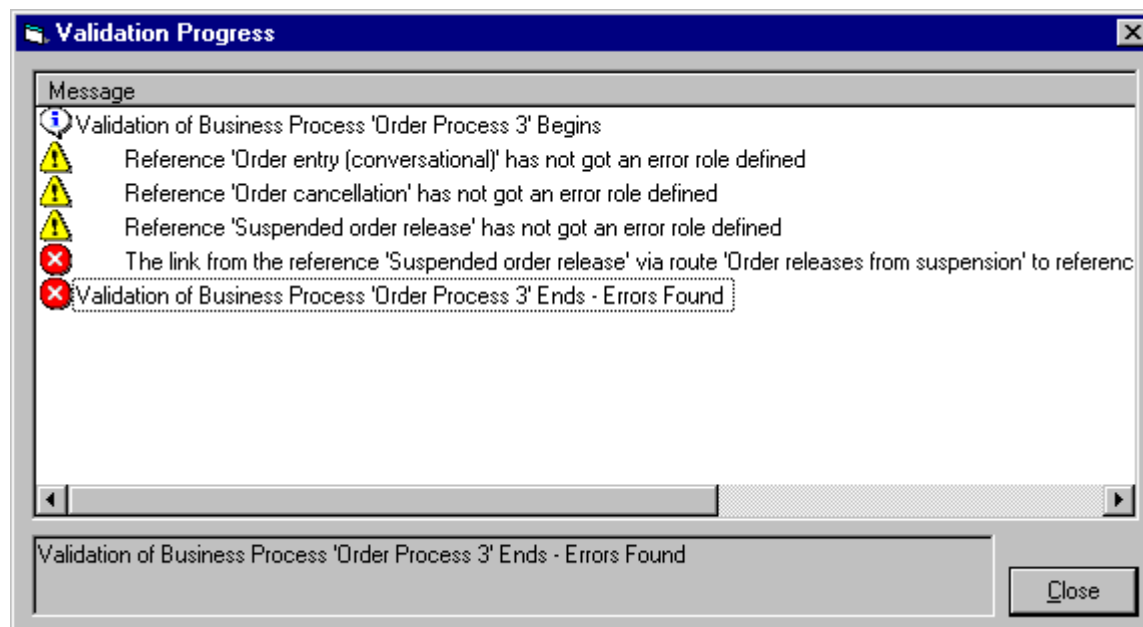
This lists any errors in the logical design.



Tip: Most errors arise because nodes are linked to activities with incorrect or missing Execution Modes.

Example Problem Solving

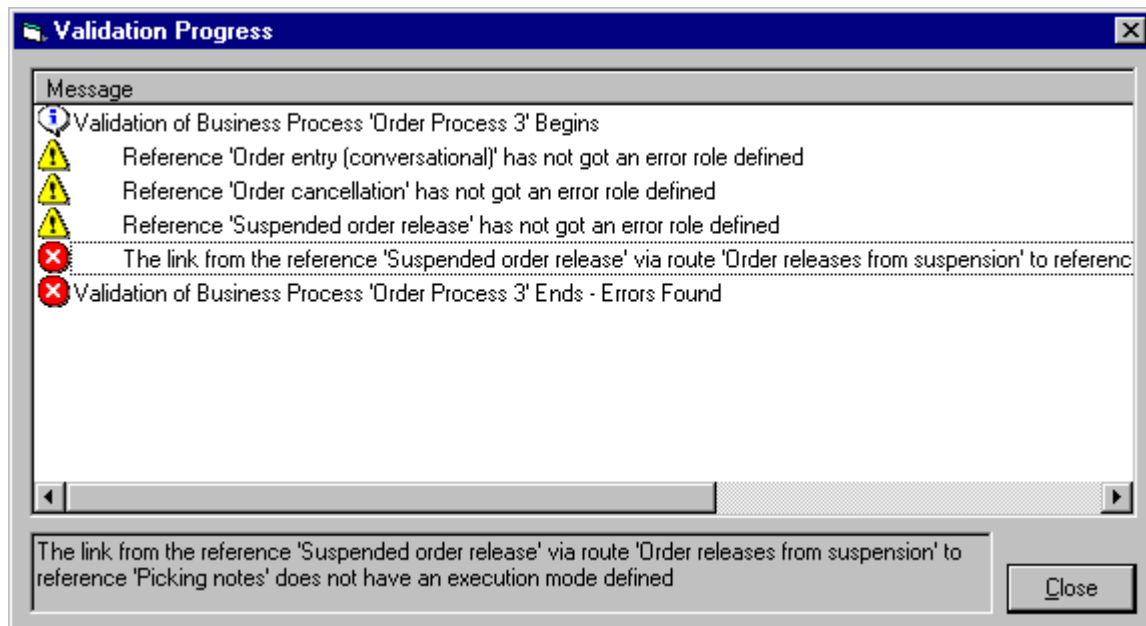
Here is a typical validation listing window.



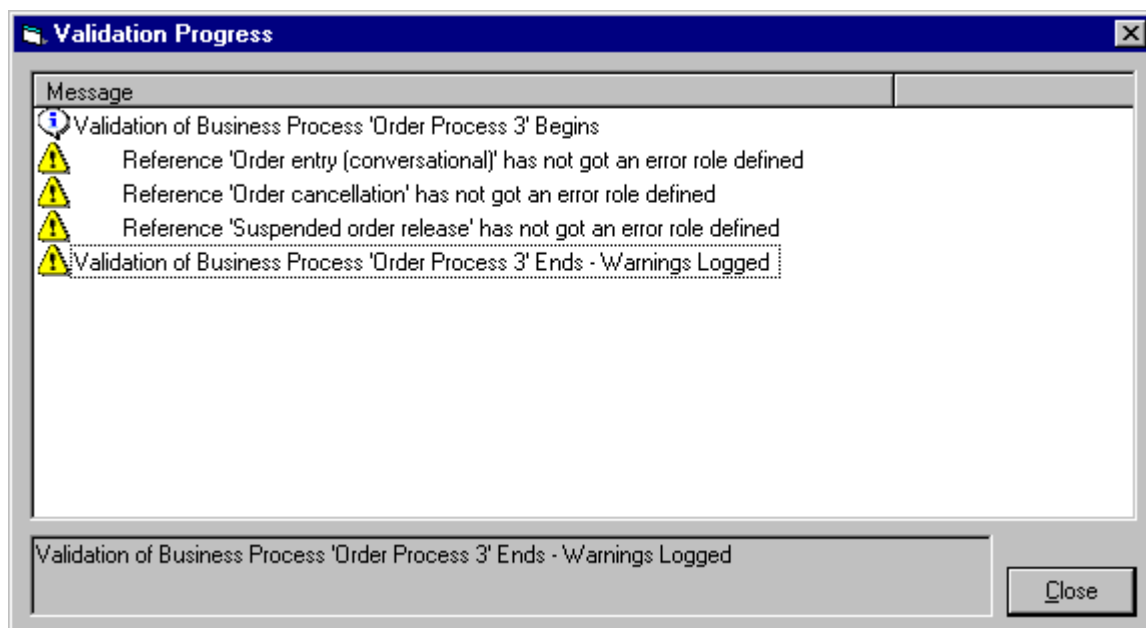
The red crosses mean errors - you must correct them.

The yellow exclamation marks are warnings - you can leave them if you wish, but they are for your information.

Select an error and the message expands in the text box at the bottom of the window:



This error arose from a missing schedule on a connection line. When it was corrected the validation showed this list of warnings below:



You will find that you get several warnings telling you that Error Roles have not been defined. Error Roles are optional so you can correct these warnings if you want to do this. You use Error Roles to define which user will receive notification of an error in their Action List. You enter this detail in the Error Role field on the Properties Work Mgt panel for each activity. Note that AE1, Ae2 and Ae3 do not specify Error Roles.

Performing A Walkthrough

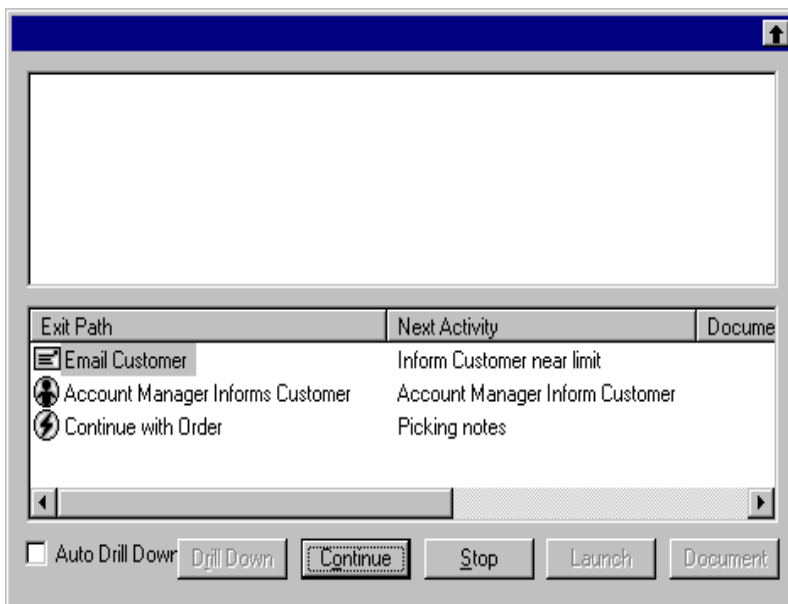
Walkthrough takes you logically through the whole process stopping at each element of the design. You can start the walkthrough anywhere.

1. Try now by selecting the Parallel Path and then select the Walkthrough button on the Toolbar. 

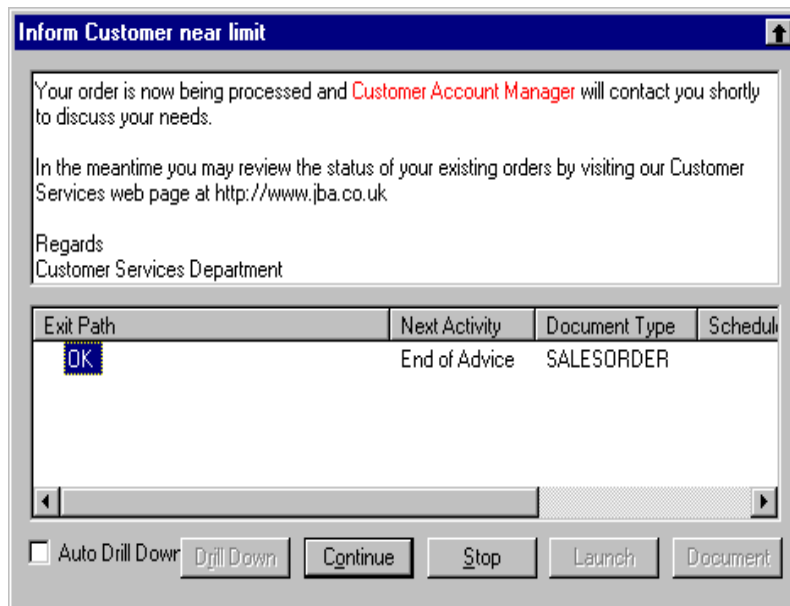


Note: Each time it hits an Assembled Activity or another Business Process it breaks it down into its lowest level components, including all its control icons. If there are Business Processes within Business Processes then all these break down too.

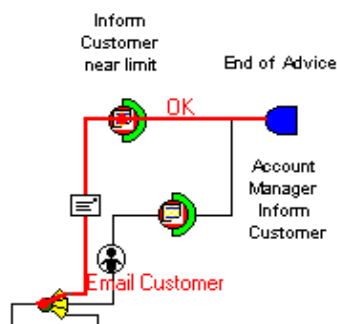
2. Each time the walkthrough stops you get a window of details. You can amend details for the component or continue. The walkthrough stops so you can select the exit from the Parallel Path as the user would. There are three options but for this example please select Email and click Continue.



3. Then it stops at the manual activity. There is only one exit from here so please click Continue.



4. Then it stops at the end.
5. This is how it should look on the Workspace. Your selected path is highlighted in red. You can now start another walkthrough.



6. When you are happy with your design you can activate it. Please refer to the help text called activating.

Checking Connections

Before you Activate please check the following:

- You need a connection to the Work Management database on the server. If you have no connection to the server, for example, if you are working on a laptop computer, then you cannot perform the activation.
- You need an ODBC data source set up using your Windows Control panel, which points to the Work Management library on the server. Refer to the documentation supplied with the communications package, for example, Client Access, which you use to communicate with your AS/400 server.

Activating

Introduction

Until now you have been saving your Business Processes on the design-time database on your PC. You cannot view them using Windows Explorer but you can see them listed in the Component Browser Area. Now you are going to Activate Order Process 3 which means uploading it to the server so that Work Management can then run it using the Work Management Engine.

Activation does the following things:

- Validate. This performs the same validation as the Validate button. If you previously validated and it failed then you will not be able to activate either.
- Compiles the configuration.
- Updates the Work Management Engine's files on the server.

The Work Management Engine checks that the Business Process has a Code. If the Business Process Code does not exist then it cannot be activated. To check this code you can right click on the Workspace and select Properties: these are the Properties relating to the entire Business Process.

This code uniquely identifies the Business Process to the Work Management Engine, whereas it is the name of the Business Process which uniquely identifies the Business Process to Constructor and to the design-time database on the PC.

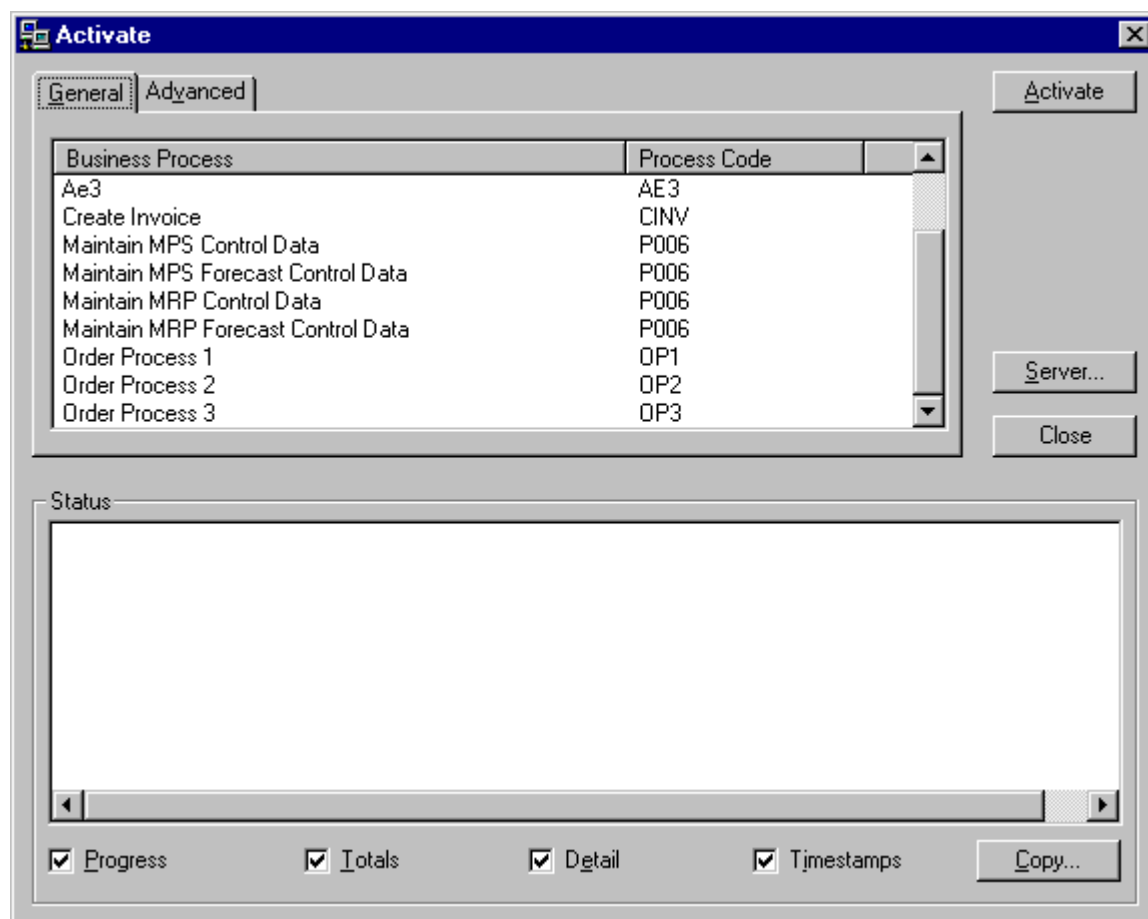
However, if you Activate the Business Process the Work Management Engine checks the Business Process Code, not the Business Process name, and saves the Business Process to the Work Management tables on the server, where the Business Process Code is now the key to the Business Process.

If the Business Process Code is new to the Work Management tables then a new file is created with a version number 001. However, if that Business Process Code already exists the Work Management Engine creates a new file for the Business Process but increments the version number. In this way all Business Processes are version controlled on the server, as identified by the Code, whereas only the latest copy of the Business Process is held on the PC and is identified by its name.

Activation Steps

This is what you do to activate:

1. Select Activate from the Work Management Menu on the Main Menu Bar. This displays the Activate Window with the General Panel selected. This panel lists all the Business Processes, saved to the design-time database on your PC, that have a Business Process Code.

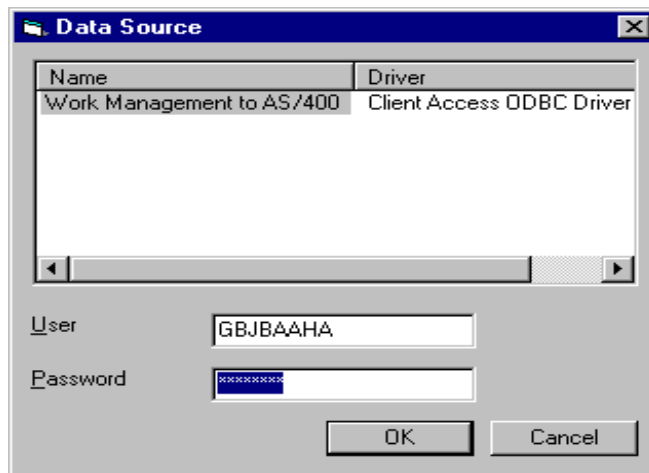


2. Scroll down the list, so that you can select Order Process 3, and then click Activate.

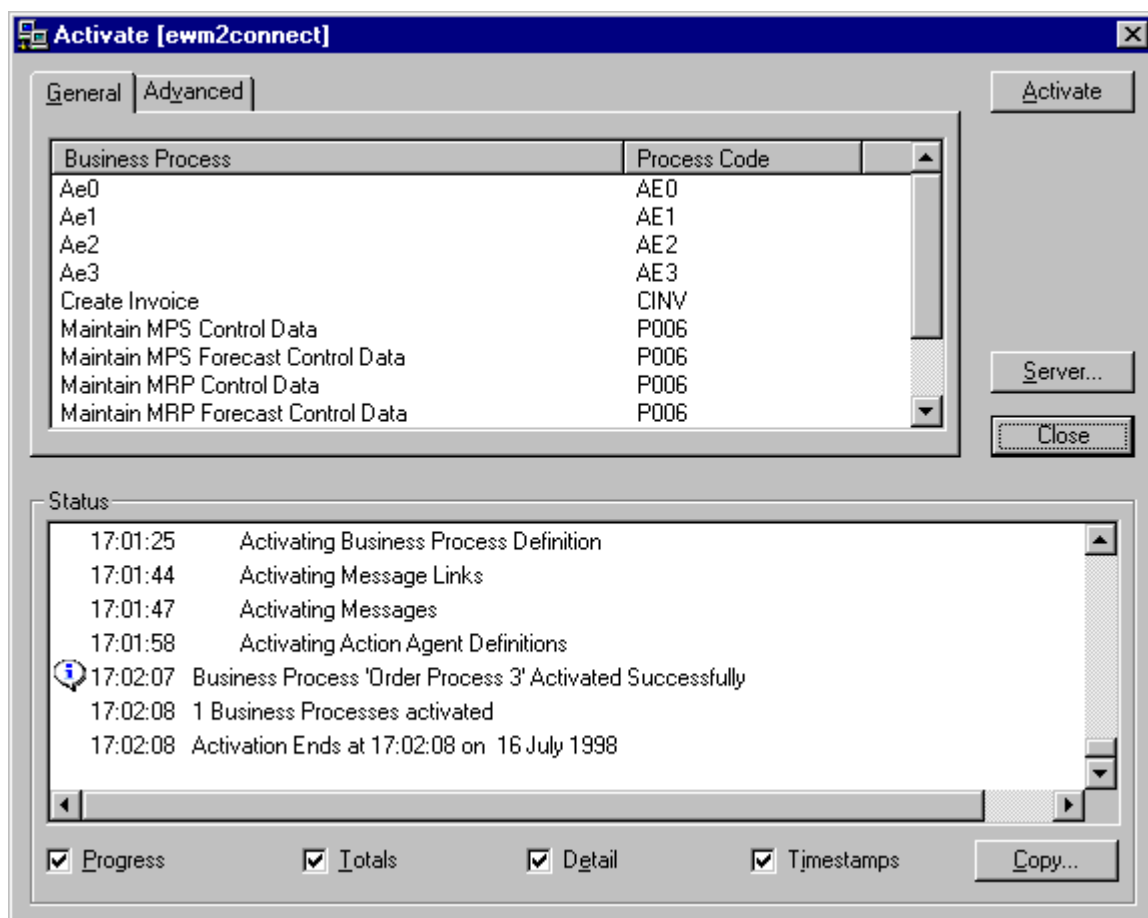
All the components associated with a Business Process are held in the design-time database on your PC with the Business Process and when you activate all these components are saved on the Work Management files on the server with the Business Process.

You can ignore the Advanced Panel for this tutorial. This is used to specifically update various components in the Work Management files.

3. You see the Validation messages in the Status box. If the validation fails with errors you will not be able to continue until you have corrected them. If the validation is successful it will say 'Validation ends'.
4. The SQL Data Sources Window is displayed for you to enter the database name. The one displayed here below is only an example. Check carefully which one you are to use.
5. You also enter a password and login.



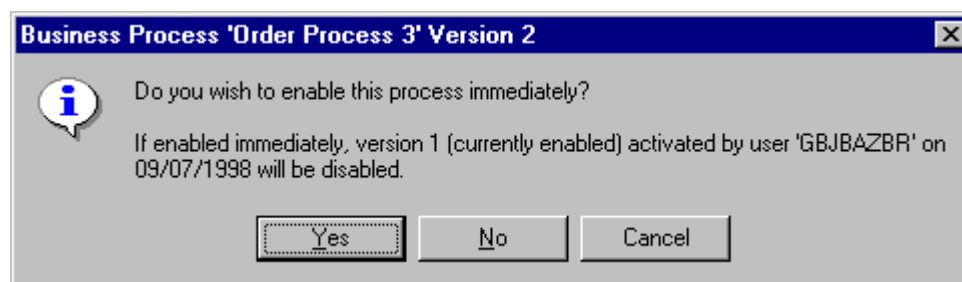
6. When the Business Process is activated you see the following window.



You have now created a version of Order Process 1 on the server.

Repeating The Process?

If you were to activate this Business Process again Constructor would recognise that there was already a previous version in existence and at the end of the Activation it would prompt you as follows:



This warning message tells you what the latest version of the Business Process is on the server, for example Version 1, and tells you which version you are trying to activate, for example, Version 2.

If you answer Yes then the previous version will be disabled and the current version will be copied to the server and enabled.

If you answer No then the previous version will remain enabled and the current version will be copied to the server and disabled.

Manage Your Business Processes

When the files are on the server you can then manage them.

Please select the Manage Activated Processes option on the Work Management Menu so that you can view the status of all the Business Processes which have been activated on the server.

You might want to view version details or delete a version. You can change the status of the Business Process to disabled or back again to enabled.

If a Business Process is disabled this means that although the Business Process exists on the server on WFP20, it is not available to the Work Management Engine for the execution of new activities. However, where documents have already been started within this Business Process they will continue until completed within this same Business Process. You can also define effectivity dates between which the Business Process will be enabled or disabled.

Note that you can also delete a Business Process using this option if you feel that you have made a mistake in activating this Business Process.

This is the end of the tutorial.